Symbiosis Institute of Technology, Nagpur

Bachelor of Technology
in
Computer Science and Engineering

"PREDICTING HUMAN STRESS LEVEL FROM PHYSIOLOGICAL SIGNALS USING MACHINE LEARNING"

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DECLARATION

I hereby declare that the project report entitled "Predicting Human Stress Level From Physiological Signals Using Machine Learning" is an original record of the work carried out by me as part of the Predictive Analytics course at Symbiosis Institute of Technology, Nagpur, towards the partial fulfilment of the requirements for the Bachelor of Technology degree in Computer Science and Engineering.

The project involves the analysis of physiological data from the WESAD and QoL_Stress datasets, the implementation and evaluation of machine learning models for binary stress classification, and a separate experiment using synthetic data. The work has been carried out under the guidance of the concerned faculty member.

I further declare that the information, research papers, datasets, and other sources used in preparing this report have been appropriately acknowledged through citations and references.

To the best of my knowledge, this report has not been submitted, in whole or in part, to any other institution for the award of a degree or academic qualification.

I understand that I am responsible for the accuracy and integrity of the work presented in this report.

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ABSTRACT

Employee Attrition Analysis and Prediction Using Logistic Regression and Risk Segmentation is an analytical project developed to identify, understand, and predict employee attrition using statistical analysis and machine learning techniques. The project analyses employee-level data consisting of **1,470 employee records**, including demographic characteristics, job-related attributes, compensation, work conditions, job satisfaction, work-life balance, business travel, and tenure-related information. The overall employee attrition rate in the dataset is **16.12%**.

The project transforms employee data into meaningful insights through data preparation, exploratory data analysis, categorical analysis, chi-square association testing, logistic regression modelling, multicollinearity analysis, odds ratio analysis, predicted probability estimation, ROC analysis, and risk segmentation. The logistic regression model evaluates the influence of factors such as overtime, department, job role, job satisfaction, job involvement, work-life balance, age, monthly income, years at company, and business travel on employee attrition.

The analysis identifies several important risk segments. Employees aged **18–25**, employees with **0–2 years of tenure**, employees with **monthly income up to 3000**, employees working **overtime**, and **Sales Representatives** show comparatively higher attrition levels. A multi-factor risk analysis further demonstrates that the likelihood of attrition increases substantially as multiple high-risk characteristics accumulate. Employees with **four or more high-risk factors** recorded an observed attrition rate of **71.83%**.

The logistic regression model achieved an **AUC of 0.8043**, indicating good discriminatory ability in distinguishing employees with higher and lower attrition risk. At the default probability threshold of 0.50, the model achieved an accuracy of **86.60%**, while threshold optimization increased sensitivity to **77.64%**, making the model more suitable for an early-warning perspective where identifying potentially at-risk employees is important.

The project demonstrates how statistical analysis and predictive modelling can be used to convert employee data into actionable HR insights. The findings can support proactive retention strategies, particularly through early-tenure support, overtime and workload management, compensation review, employee engagement initiatives, and targeted interventions for employees exhibiting multiple attrition-risk characteristics.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Stress is a physiological and psychological response to situations that place demands on an individual. It can be accompanied by changes in cardiovascular activity, electrodermal responses, respiration, and other physiological processes. These changes have motivated research into the use of wearable sensing technologies for identifying physiological patterns associated with stress. However, physiological measurements do not provide a direct or definitive measure of a person's psychological state, and their interpretation depends on the individual, experimental conditions, and measurement methods [1], [2]. Traditional stress assessment methods often involve questionnaires, interviews, and subjective self-reports. Although these methods provide valuable information about perceived stress, they may not capture changes in physiological activity continuously. Wearable devices offer an alternative by enabling the collection of physiological measurements that can be analysed using statistical techniques and machine learning. Research datasets such as Wearable Stress and Affect Detection (WESAD) have supported the development and evaluation of machine learning approaches for stress recognition using multimodal physiological signals [1].

Physiological measurements provide complementary information about the body's response to different conditions. Electrocardiogram (ECG) signals represent the electrical activity of the heart, while electrodermal activity (EDA) reflects changes in skin conductance associated with sweat gland activity. Respiration and body temperature provide additional information about physiological responses. Analysing features derived from these signals may help distinguish between experimentally labelled baseline and stress conditions. Nevertheless, differences in individual physiology, sensor characteristics, and data collection protocols can make reliable classification challenging [2], [3]. Machine learning provides a systematic approach to identifying patterns in physiological data and using them to predict labelled outcomes. In this approach, physiological features are supplied to a classification algorithm, which learns relationships between the input measurements and the corresponding experimental labels. The resulting model can then generate predictions for observations that were not used during training. However, a model's performance on previously observed individuals may not represent its performance on new individuals. Subject-specific physiological characteristics can influence the results, making participant-independent evaluation important when developing wearable stress classification systems [3], [4].

The project, titled Predicting Human Stress Level From Physiological Signals Using Machine Learning, investigates the application of supervised machine learning to distinguish baseline or relaxation observations from acute stress observations. The study uses two empirical datasets: WESAD, which provides physiological recordings collected under controlled laboratory conditions, and QoL_Stress, which provides physiological measurements associated with controlled relaxation and stress sessions using consumer wearable technology [1], [2]. The datasets are analysed separately because their recording structures, feature representations, and observation counts differ.

Three classification algorithms are evaluated: Logistic Regression, Random Forest, and HistGradientBoosting. Their performance is assessed using Leave-One-Subject-Out (LOSO) cross-validation, in which observations belonging to one participant are held out for testing while the remaining participants are used for training. This evaluation strategy is intended to reduce participant-level information leakage and provide a more meaningful assessment of performance on unseen individuals [3], [4]. The models are compared using balanced accuracy, macro F1-score, precision, recall, the area under the receiver operating characteristic curve (ROC-AUC), and the area under the precision-recall curve (PR-AUC), where applicable.

In addition to the two empirical datasets, the project includes a separate experiment using synthetic data. This experiment examines classification performance under simulated conditions and is interpreted independently of the results obtained from real physiological recordings. By comparing the models across the empirical datasets and clearly separating the synthetic experiment, the study aims to demonstrate the importance of data characteristics, participant-independent validation, and appropriate evaluation metrics in physiological stress prediction.

The project is an academic predictive analytics investigation rather than a clinically validated diagnostic system. Its purpose is to evaluate machine learning methods on the selected datasets, interpret their recorded performance, and identify limitations that should be addressed in future research.

1.2 Background and Motivation

The use of physiological signals for stress classification has become an important area of research in wearable sensing and machine learning. Stress can be associated with changes in cardiovascular activity and electrodermal responses, which can be measured using sensors available in wearable devices. Electrocardiography (ECG) records the electrical activity of the heart, while electrodermal activity (EDA) measures changes in the electrical conductance of the skin. Heart rate and heart rate variability (HRV) can also provide information about cardiovascular responses. These measurements can be analysed to identify patterns associated with experimentally labelled stress conditions. However, physiological responses vary between individuals, and the same signal may be influenced by factors other than stress [3].

The development of publicly available physiological datasets has supported research into automated stress classification. The WESAD dataset introduced by Schmidt et al. [1] contains multimodal physiological recordings collected from 15 participants under controlled baseline, stress, and amusement conditions. It provides a basis for studying the relationship between physiological measurements and experimentally labelled states. The QoL_Stress dataset provides another experimental setting, containing physiological recordings associated with relaxation and stress-induction phases. Its measurements include ECG, EDA, and heart rate collected using wearable technology [2]. These datasets differ in their recording procedures, available features, and observation structures, making separate evaluation appropriate for this project.

Machine learning offers a method for identifying relationships between physiological features and predefined class labels. Instead of relying on a manually specified rule for every physiological measurement, a classification algorithm learns patterns from labelled examples. The learned model can then predict the class of new observations. Nevertheless, a model's performance depends on the quality and characteristics of the training data, the selected features, and the evaluation strategy. A systematic review by Vos et al. [4] highlights the importance of generalization when developing machine learning approaches for wearable stress monitoring.

One key motivation for this project is the challenge of evaluating models on participants whose observations were not used during training. Physiological characteristics can differ between individuals, potentially causing a model to perform differently on unseen participants. If observations from the same participant appear in both training and testing sets, the evaluation may not adequately represent performance on a new individual. Participant-independent validation is therefore relevant when assessing the generalization of stress classification models [4], [5].

To investigate this issue, the project evaluates Logistic Regression, Random Forest, and HistGradientBoosting using the implemented physiological feature datasets. For WESAD, the analysis uses a binary classification task distinguishing baseline observations from stress

observations. For QoL_Stress, the task distinguishes relaxation from stress observations. Leave-One-Subject-Out (LOSO) cross-validation is used for the WESAD experiment to evaluate predictions for participants held out from model training. The QoL_Stress results are analysed separately, taking into account the limitations of its participant-level evaluation and the availability of class observations in individual folds.

The motivation for comparing multiple algorithms is to understand how different classification approaches perform under the same experimental conditions. Logistic Regression provides a linear classification baseline, Random Forest combines predictions from multiple decision trees, and HistGradientBoosting builds an ensemble of trees through sequential boosting. Their performance is compared using balanced accuracy, macro F1-score, stress-class recall, ROC-AUC, and PR-AUC, as appropriate to each evaluation.

The implemented results illustrate why this comparison is necessary. In the WESAD experiment, HistGradientBoosting achieved a balanced accuracy of 0.9311 and a macro F1-score of 0.9384 under pooled LOSO evaluation. In the QoL_Stress experiment, Logistic Regression achieved the highest balanced accuracy among the three evaluated models, at 0.6203. These results differ across the two datasets and demonstrate that model performance should be interpreted in relation to the dataset, experimental protocol, and evaluation method rather than assumed to transfer directly between settings. The project also includes a separate synthetic-data experiment. This experiment evaluates the classification algorithms on simulated observations and provides an additional controlled comparison. Its results are not combined with the empirical results from WESAD or QoL_Stress because synthetic observations do not represent additional human participants or real physiological recordings. The overall motivation of this study is to evaluate machine learning methods for physiological stress classification while examining the effects of dataset characteristics, model choice, and participant-independent validation. The findings are intended to contribute to an understanding of the strengths and limitations of the implemented approaches. The project does not establish clinical validity or demonstrate reliable stress detection in unrestricted real-world environments.

1.3 Problem Statement

Physiological signals such as electrocardiography (ECG), heart rate variability (HRV), and electrodermal activity (EDA) can provide information about bodily responses associated with stress. However, identifying stress from these measurements is challenging because physiological patterns differ between individuals and may also be influenced by factors unrelated to stress. Consequently, developing a machine learning model that classifies physiological observations accurately requires appropriate feature preparation, reliable evaluation, and careful interpretation of the results [3], [4].

A major challenge in wearable stress classification is determining whether a trained model can classify observations from participants whose data were not used during training. When training and testing observations contain information from the same participants, the evaluation may not adequately reflect performance on previously unseen individuals. Research on generalizable stress monitoring and cross-dataset evaluation highlights the importance of investigating model generalization rather than relying exclusively on classification performance within a single experimental setting [4], [5].

This project addresses the problem of classifying physiological observations into two categories using supervised machine learning. The WESAD dataset is used to distinguish baseline observations from stress observations, while the QoL_Stress dataset is used to distinguish relaxation

observations from stress observations. These datasets are evaluated independently because their recording protocols, feature representations, and observation structures differ [1], [2].

The study compares Logistic Regression, Random Forest, and HistGradientBoosting to determine how their classification performance differs across the selected datasets. For WESAD, Leave-One-Subject-Out (LOSO) cross-validation is used to evaluate the models on participants excluded from the corresponding training fold. The QoL_Stress experiment is evaluated separately, with its results interpreted in light of the available observations and participant-level validation limitations.

The central problem can be expressed mathematically as learning a classification function:

$$\hat{y} = f(\mathbf{x}; \theta)$$

where:

- $\mathbf{x}$ represents the physiological features supplied to the model.
- f represents the selected machine learning algorithm.
- θ represents the parameters learned during training.
- $\hat{y}$ represents the predicted physiological state.

The predicted category is interpreted according to the dataset used: baseline or stress for WESAD, and relaxation or stress for QoL_Stress.

The models are assessed using balanced accuracy, macro F1-score, precision, recall, ROC-AUC, and PR-AUC, where applicable. The project also includes a separate synthetic-data experiment, whose results are reported independently from those obtained using empirical physiological recordings.

Therefore, the problem addressed by this project is to evaluate the performance of selected machine learning algorithms for binary physiological stress classification, examine their performance under the implemented validation procedures, and identify the limitations that affect the interpretation and generalizability of their predictions. The study does not claim to establish a clinically validated stress detection system.

1.4 Aim and Objectives

1.4.1 Aim

The aim of this project is to develop and evaluate supervised machine learning models for binary stress classification using physiological features derived from the WESAD and QoL_Stress datasets. The study compares Logistic Regression, Random Forest, and HistGradientBoosting to examine their classification performance under the implemented evaluation procedures. It also investigates the importance of participant-independent validation in assessing model performance and analyses a separate synthetic-data experiment without combining its results with those obtained from empirical physiological recordings.

1.4.2 Objectives

The specific objectives of the project are:

1. To study and prepare the WESAD and QoL_Stress datasets for binary classification of baseline or relaxation and stress observations, according to the labels available in each dataset.
2. To perform the required data preprocessing and feature preparation while excluding identifiers, target-derived information, and other features that could introduce data leakage.
3. To implement Logistic Regression, Random Forest, and HistGradientBoosting for physiological stress classification.
4. To evaluate the models using appropriate performance metrics, including balanced accuracy, macro F1-score, precision, recall, ROC-AUC, and PR-AUC, where applicable.
5. To apply Leave-One-Subject-Out cross-validation in the WESAD experiment to assess classification performance on participants excluded from the corresponding training fold.
6. To compare the performance of the three classification algorithms separately on WESAD and QoL_Stress, accounting for differences in dataset characteristics and evaluation limitations.
7. To conduct a separate classification experiment using synthetic data and report its findings independently of the empirical dataset results.
8. To interpret the experimental findings, identify limitations affecting model performance and generalization, and suggest directions for future research in physiological stress classification.

1.5 Research Questions

This project investigates the application of supervised machine learning to physiological stress classification using the WESAD and QoL_Stress datasets. The following research questions guide the experimental analysis.

RQ1. How effectively can Logistic Regression, Random Forest, and HistGradientBoosting distinguish baseline or relaxation observations from stress observations using physiological features?

RQ2. How does the classification performance of the three machine learning algorithms differ between the WESAD and QoL_Stress datasets?

RQ3. How well do the trained models generalize to unseen participants when evaluated using Leave-One-Subject-Out cross-validation in the WESAD experiment?

RQ4. How do the selected classification algorithms compare in terms of balanced accuracy, macro F1-score, precision, recall, ROC-AUC, and PR-AUC?

RQ5. How do dataset characteristics, including sample size, feature representation, class distribution, and recording conditions, affect the interpretation of the classification results?

RQ6. How do the classification results obtained from synthetic data compare across the three algorithms, and why must these results be interpreted separately from those obtained using empirical physiological recordings?

These research questions focus on the implemented experiments and their measurable outcomes. They do not assume that a model performing well on one dataset will achieve equivalent performance on another dataset or in real-world environments.

1.6 Scope of the Project

The scope of this project is limited to the development and evaluation of supervised machine learning models for binary stress classification using physiological features. The study focuses on the selected datasets, implemented preprocessing procedures, classification algorithms, and evaluation methods.

1.6.1 Dataset Scope

The project uses two empirical datasets:

1. WESAD: The dataset is used to classify physiological observations into baseline and stress categories. The implemented experiment uses 2,575 windows with 144 physiological predictors from 15 participants. The amusement class is excluded from the binary classification task.

2. QoL_Stress: The dataset is used to distinguish relaxation and stress observations. The implemented dataset contains 124 session-level observations from 64 participants. The analysis uses physiological features derived from ECG and EDA measurements, with additional demographic and questionnaire variables considered separately in the feature-ablation experiment.

The datasets are processed and evaluated independently because they differ in their data collection procedures, feature representations, and observation structures.

1.6.2 Machine Learning Scope

The project evaluates three supervised classification algorithms:

- Logistic Regression
- Random Forest
- HistGradientBoosting

The study compares these algorithms using the physiological features prepared from the selected datasets. It does not aim to develop a new machine learning algorithm or establish a universally optimal classifier for stress detection.

1.6.3 Data Preprocessing and Feature Preparation

The project includes data preparation steps required for reliable model evaluation. These include selecting appropriate predictors, handling missing values where required, excluding identifiers and target-derived information, and applying feature scaling within the training pipeline where appropriate.

For WESAD, the processed observations are represented as overlapping 60-second windows with a 10-second stride. The preprocessing and validation procedures are designed to reduce information leakage between training and testing data.

1.6.4 Model Evaluation Scope

The WESAD experiment uses Leave-One-Subject-Out (LOSO) cross-validation, in which one participant is held out for testing while the remaining participants are used for training. This provides a participant-independent evaluation of the implemented models.

The QoL_Stress experiment is analysed separately. Its pooled out-of-fold results are reported independently, and limitations arising from single-class validation folds are considered when interpreting fold-level metrics.

The evaluation includes balanced accuracy, macro F1-score, precision, recall, ROC-AUC, and PR-AUC, where applicable. These metrics are used to compare classification performance rather than to establish clinical effectiveness.

1.6.5 Synthetic-Data Experiment

The project includes a separate experiment involving 2,500 synthetic observations, 50 simulated subjects, and 10 features. The three classification algorithms are evaluated on this dataset to examine their performance under simulated conditions.

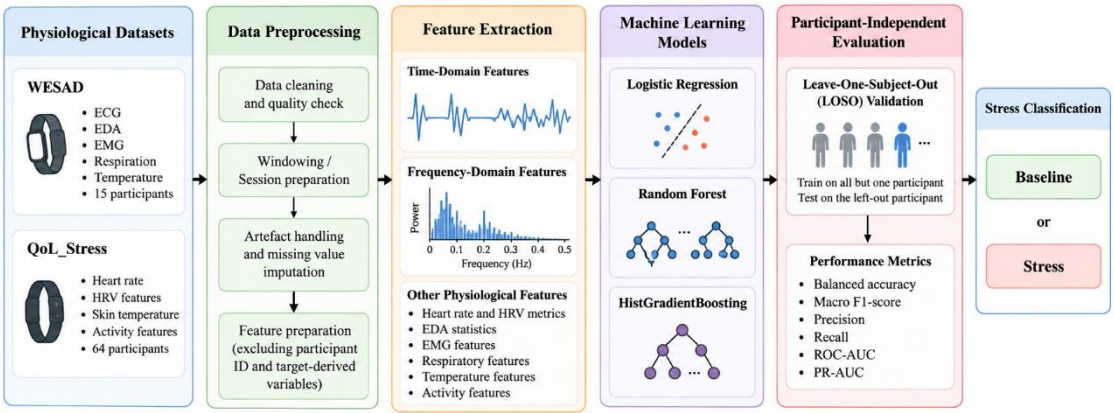
The synthetic experiment is not treated as an extension of the empirical datasets. Its results are reported separately and are not used as evidence of performance on real human physiological data.

1.6.6 Limitations and Exclusions

The project is restricted to the datasets, features, algorithms, and evaluation procedures implemented during the study. It does not include prospective data collection from new participants, real-time deployment on wearable devices, clinical validation, or testing in unrestricted everyday environments.

Furthermore, the classification labels represent the conditions recorded in the respective datasets. The models predict these predefined categories and do not directly measure psychological stress or establish a medical diagnosis.

The findings are therefore interpreted as an experimental evaluation of machine learning methods for physiological stress classification within the scope of the selected datasets. Further research using independent datasets, additional participants, and real-world validation would be required before considering practical clinical or continuous-monitoring applications.



1.7 Significance of the Study

The significance of this project lies in the application and comparative evaluation of machine learning algorithms for physiological stress classification. By using the WESAD and QoL_Stress datasets, the study examines how selected classification methods perform under different experimental settings. Previous research has highlighted the potential of physiological signals for stress monitoring, while also identifying generalization as an important challenge in wearable stress detection [3], [4].

First, the project provides a comparative evaluation of Logistic Regression, Random Forest, and HistGradientBoosting. Applying these algorithms to the same prepared datasets allows their classification performance to be examined using consistent evaluation metrics within each experiment. This comparison helps identify differences in predictive performance rather than assuming that one algorithm is suitable for every dataset.

Second, the project emphasizes participant-independent evaluation through Leave-One-Subject-Out (LOSO) cross-validation in the WESAD experiment. Holding out one participant during each training and testing cycle provides a more appropriate assessment of performance on participants excluded from model training. This is relevant because individual physiological differences can affect the generalization of wearable stress classification models [4], [5].

Third, the separate evaluation of WESAD and QoL_Stress highlights the importance of dataset characteristics when interpreting machine learning results. Differences in recording protocols, features, sample structures, and experimental conditions can influence classification performance. Evaluating the datasets independently avoids treating their results as directly interchangeable and helps identify the limitations of conclusions drawn from a single experimental setting.

Fourth, the project uses multiple evaluation metrics, including balanced accuracy, macro F1-score, precision, recall, ROC-AUC, and PR-AUC, where applicable. Considering multiple metrics provides a broader assessment of classification performance, particularly when class distributions are unequal. It also helps distinguish overall classification performance from the model's ability to identify stress observations.

Finally, the separate synthetic-data experiment provides an additional setting in which to compare the three algorithms. Keeping these results separate from the empirical dataset results maintains a clear distinction between simulated observations and physiological recordings collected from human participants.

Overall, the study contributes an implementation-based comparison of machine learning approaches for binary physiological stress classification. Its findings can support further investigation into participant-independent evaluation, dataset-specific model performance, and the limitations of applying laboratory-based results to real-world stress monitoring. However, the project does not establish clinical validity or demonstrate that the evaluated models are suitable for direct medical use.

1.8 Organization of the Report

This report is organized into six chapters, following the required project report structure.

Chapter 1: Introduction presents the background and motivation for physiological stress classification, the problem statement, aim and objectives, research questions, scope, and significance of the study.

Chapter 2: Literature Review examines existing research on physiological stress indicators, wearable sensing, machine learning approaches for stress classification, and the challenges associated with model generalization. It establishes the research context for the implemented project.

Chapter 3: Workflow describes the methodology followed in the project. It covers the WESAD and QoL_Stress datasets, data preprocessing, feature preparation, classification algorithms, validation procedures, evaluation metrics, and the separate synthetic-data experiment.

Chapter 4: Results present the experimental findings obtained from the implemented models. It compares Logistic Regression, Random Forest, and HistGradientBoosting across the two empirical datasets and reports the synthetic-data results separately. Appropriate tables and project-generated figures are used to present the findings.

Chapter 5: Conclusion summarizes the main findings in relation to the research questions and project objectives. It discusses the limitations of the experiments and identifies possible directions for future research.

Chapter 6: References lists the academic papers, dataset documentation, and other sources cited throughout the report in IEEE referencing style.

CHAPTER 2: LITERATURE REVIEW

2.1 Physiological Stress and Wearable Sensors

Stress is associated with changes in the body's physiological activity as it responds to internal or external demands. These changes can involve the autonomic nervous system, cardiovascular activity, and electrodermal responses. Physiological measurements therefore provide useful information for investigating stress-related states. However, no single physiological measurement is a direct or universally reliable indicator of stress, because similar responses may occur during physical activity, emotional excitement, changes in posture, or other conditions [3].

Wearable devices have made physiological monitoring more accessible by collecting measurements outside conventional laboratory equipment. Depending on the device, these measurements may include electrocardiography (ECG), heart rate, heart-rate variability (HRV), electrodermal activity (EDA), skin temperature, respiration, and movement. ECG records electrical activity associated with the heart, while HRV features describe variations in the intervals between successive heartbeats. EDA measures changes in the skin's electrical conductance associated with sweat-gland activity and sympathetic nervous system responses. These measurements can provide complementary information about physiological arousal [3].

Machine learning can analyse multiple physiological features simultaneously and learn patterns associated with experimentally labelled conditions. Instead of depending on a single threshold, a classifier can use combinations of measurements to distinguish between classes represented in its training data. Nevertheless, the presence of a physiological pattern does not establish that a person is experiencing stress in every context. Model predictions depend on the quality of the measurements, the reliability of the labels, the characteristics of the participants, and the conditions under which the data were collected.

A major challenge is individual variability. Two people exposed to the same experimental condition may exhibit different physiological responses. In addition, the same individual may show different responses depending on activity, health, environment, or the intensity of the stimulus. Vos et al. [4] identify generalization as an important challenge in wearable stress monitoring, emphasizing that performance on one group of participants may not necessarily transfer to unseen individuals. This makes the validation strategy as important as the choice of classifier.

The present study examines this issue through two datasets with different measurement characteristics: WESAD, which provides continuous physiological recordings collected using research-oriented wearable sensors, and QoL_Stress, which includes episodic physiological measurements from a consumer fitness tracker. Analysing the datasets separately allows the study to investigate how classification performance differs across these two experimental settings. The

objective is to evaluate the predictive capability of the implemented models, not to establish a clinically validated stress assessment system.

2.2 WESAD Dataset and Its Role in Stress Detection

2.2.1 Overview of the WESAD Dataset

The Wearable Stress and Affect Detection (WESAD) dataset was introduced by Schmidt et al. [1] as a multimodal dataset for research on stress and affect recognition using wearable sensors. It has become a useful benchmark for studying physiological patterns associated with experimentally induced states. The dataset contains recordings from 15 participants who completed a controlled protocol involving baseline, stress, and amusement conditions.

An important contribution of WESAD is the availability of multiple physiological modalities collected during the same experimental protocol. This enables researchers to investigate whether information from different sensors can support the classification of labelled physiological states. It also allows comparisons between features derived from cardiovascular, electrodermal, respiratory, temperature, and movement measurements.

The dataset is particularly relevant to supervised machine learning because its experimental conditions provide labels that can be associated with physiological observations. However, these labels represent the states defined by the experimental protocol. They should not be interpreted as universal labels for every real-world experience of stress or as clinical diagnoses.

In the present study, WESAD serves as the laboratory-oriented benchmark. The prepared feature matrix contains 2,575 sixty-second windows from 15 participants and 144 physiological predictors. The implemented binary classification task distinguishes baseline from stress windows. The amusement class is excluded from this specific experiment so that the model comparison focuses on the distinction between baseline and stress.

2.2.2 Experimental Conditions and Physiological Modalities

WESAD combines chest-worn and wrist-worn sensing. The chest-worn RespiBAN device records several physiological signals, including ECG, respiration, electromyography, EDA, temperature, and acceleration. The wrist-worn Empatica E4 provides measurements such as blood-volume pulse, EDA, temperature, and acceleration. These modalities differ in their physiological meaning, sampling characteristics, and sensitivity to movement and sensor placement [1].

The use of multiple modalities is valuable because physiological stress responses are not restricted to a single signal. Cardiac features may capture changes in heart activity, while EDA features may reflect changes in sympathetic arousal. Respiration and temperature can provide additional context. Combining features can therefore allow a classifier to identify patterns that may not be evident from an individual measurement alone.

However, a multimodal dataset also introduces analytical challenges. Signals may have different sampling rates, scales, and noise characteristics. They must be aligned appropriately before features from the same observation can be analysed together. Poor alignment, incorrect label mapping, or features that indirectly reveal the target can produce misleading evaluation results.

In the implemented WESAD experiment, the continuous recordings were represented as sixty-second windows with a ten-second stride. This creates an overlap of approximately 83.3% between adjacent windows. Although overlapping windows provide more observations for model development, they are not statistically independent because neighbouring windows may contain

much of the same underlying signal. A random window-level train-test split could consequently allow very similar observations from the same participant to appear in both sets.

For this reason, the present study uses Leave-One-Subject-Out (LOSO) cross-validation. Each evaluation fold holds out one participant for testing and trains the model using the remaining participants. This evaluates whether a model can classify observations from an individual whose data were not used to fit that fold's model.

2.2.3 Relevance of WESAD to the Present Study

WESAD provides a suitable starting point for comparing supervised classification methods under a controlled experimental protocol. Its continuous recordings support window-level feature analysis, while the availability of participant identifiers enables participant-independent validation. These characteristics make it useful for studying both predictive performance and the risk of participant-level data leakage.

The present study compares Logistic Regression, Random Forest, and HistGradientBoosting on the prepared WESAD feature matrix. The three models represent different learning approaches: a regularized linear classifier, a tree-based ensemble, and a gradient-boosted tree method. Comparing them under the same validation framework helps isolate differences in predictive performance associated with the chosen algorithms.

The dataset also provides a reference point for interpreting results from QoL_Stress. However, the two datasets differ in sensor configuration, recording duration, sampling characteristics, feature construction, and the number of participants and observations. Their results must therefore be interpreted as outcomes from separate experimental tracks rather than as a controlled comparison in which dataset type is the only changing factor.

WESAD's main value in this project is consequently twofold: it provides a benchmark for evaluating the selected classifiers and offers a setting in which participant-independent model evaluation can be demonstrated. Its controlled nature also defines a limitation: successful classification on WESAD does not, by itself, establish equivalent performance with consumer wearables or in everyday environments.

2.3 QoL_Stress Dataset and Consumer Wearable Measurements

2.3.1 Overview of the QoL_Stress Dataset

QoL_Stress is a multimodal dataset designed to investigate physiological and self-reported responses to relaxation and acute stress conditions. The dataset is available through PhysioNet and is described by Calvo, Cejudo, and Martín Andonegui [2]. It is relevant to wearable stress research because it includes measurements collected using a consumer fitness tracker rather than relying exclusively on research-oriented laboratory sensors.

The distinction between laboratory-oriented and consumer-wearable measurements is important. Research-grade systems may provide continuous recordings from several sensors, whereas consumer devices may provide shorter recordings, device-processed measurements, or data collected only when a user initiates a measurement. These differences affect the amount and temporal resolution of information available to a machine-learning model.

In the present study, QoL_Stress is represented by 124 session-level observations from 64 unique participants. The primary feature matrix contains 25 physiological predictors, consisting of 15 ECG and HRV features and 10 EDA features. Unlike WESAD, where the prepared observations are

generated from overlapping continuous windows, the QoL_Stress observations represent individual experimental sessions.

The dataset therefore provides a complementary evaluation setting. It allows the project to examine whether the same broad families of machine-learning classifiers can distinguish relaxation and stress labels when the available physiological information is more episodic and the number of labelled observations is comparatively limited.

2.3.2 Physiological Measurements and Recording Protocol

The QoL_Stress measurements used in this project include ECG and EDA features derived from recordings made with the Fitbit Charge 6. The ECG data consist of short Lead-I waveform strips sampled at 250 Hz, while the EDA recordings are approximately 2.7 minutes long and sampled at 1 Hz. These measurements differ substantially from the continuous, multi-sensor recordings available in WESAD [1], [2].

ECG-derived features can describe aspects of cardiac activity, including heart rate and variations between successive beats. EDA-derived features describe the level and variation of skin conductance, which can provide information about sympathetic arousal. The combination of cardiac and electrodermal features offers two complementary perspectives on physiological response. Nevertheless, neither modality is specific to psychological stress, and measurements may also be affected by recording conditions, motion, sensor contact, and individual differences [3].

The episodic nature of QoL_Stress creates a different feature-engineering problem from that of WESAD. In WESAD, features are calculated from repeated fixed-length windows within continuous recordings. In QoL_Stress, the model operates on session-level features derived from shorter recordings. As a result, the number and nature of the available observations differ between the two datasets.

Another important consideration is the alignment of measurements with experimental labels. Background heart-rate telemetry without reliable condition timestamps cannot automatically be assigned to relaxation or stress sessions. Doing so could introduce label ambiguity or incorrect target assignments. Accordingly, the project's preprocessing workflow excludes unaligned background telemetry from the supervised session-level feature matrix.

The modelling workflow also excludes participant identifiers and the experimental condition field from the predictor variables. Post-condition state-anxiety scores are excluded because they are measured after the experimental condition and could reveal information associated with the target. This is important because a model should learn from permitted physiological predictors rather than from variables that directly identify the participant or encode the answer.

2.3.3 Relevance of QoL_Stress to the Present Study

QoL_Stress allows the project to examine a setting in which the number of observations is relatively small, measurements are episodic, and the available physiological feature space is narrower than that used in the WESAD experiment. These characteristics make it useful for assessing the limitations of wearable stress classification beyond a continuous laboratory dataset.

The project evaluates the same three classifier families on QoL_Stress: Logistic Regression, Random Forest, and HistGradientBoosting. The primary comparison uses physiological features only. A secondary feature-ablation experiment investigates the effect of adding demographic and baseline trait-stress variables. These additional variables are analysed separately so that their contribution can be distinguished from that of the primary physiological feature set.

The dataset requires cautious interpretation. Although the project includes 124 session observations, those observations come from only 64 participants, and some participants contribute measurements for both conditions. The number of sessions therefore does not represent 124 independent individuals. Participant-aware validation is necessary to avoid placing observations from the same individual in both training and testing data.

The use of QoL_Stress does not establish that the model is ready for continuous monitoring through a commercial smartwatch. The recordings are collected under controlled relaxation and acute-stress protocols, and the feature matrix is session-based. The experiment instead provides evidence about the performance of the tested models on this particular dataset and illustrates the practical difficulties of classification with episodic consumer-wearable measurements.

2.4 Machine Learning Approaches for Stress Classification

Supervised machine learning frames stress classification as a prediction task in which a model learns a relationship between input features and known class labels. After training, the model predicts a class for an observation whose label is not supplied as an input. In physiological stress research, the input may consist of statistical summaries of ECG, HRV, EDA, respiration, temperature, or other signals.

The usefulness of a model depends not only on its algorithm but also on the quality of the features, the representativeness of the training observations, the reliability of the labels, and the evaluation procedure. A complex model can perform well on observations similar to its training data but fail to generalize to new participants. A simpler model may be less flexible but easier to interpret and evaluate. Reviews of wearable stress monitoring identify generalization and evaluation design as important considerations in the development of reliable systems [4], [7].

The present study compares three classifiers that represent different modelling strategies. All are evaluated separately on WESAD and QoL_Stress using the same general participant-independent validation principle. This provides a practical comparison of linear modelling, bagged decision trees, and gradient-boosted decision trees within the scope of the implemented project.

2.4.1 Logistic Regression

Logistic Regression is a supervised classification algorithm that estimates the probability of an observation belonging to a class. For binary classification, it applies the logistic function to a weighted combination of input features. The estimated probability can then be converted into a predicted class using a decision threshold.

The model can be expressed as:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + \exp[-(\beta_0 + \boldsymbol{\beta}^T \mathbf{x})]}$$

where $\mathbf{x}$ represents the input feature vector, β_0 represents the feature coefficients, and β_0 is the intercept. The predicted probability represents the model's estimated likelihood of the positive class under the fitted model; it should not automatically be interpreted as a calibrated probability of a person's true psychological state.

Logistic Regression is useful as a baseline because it is computationally efficient and models an approximately linear relationship between the input features and the log-odds of the outcome. With suitable regularization, it can reduce the effect of excessively large coefficients and help control

model complexity. Feature scaling is particularly relevant when predictors have different numerical ranges.

A limitation is that Logistic Regression does not automatically capture complex nonlinear relationships between features unless those relationships are represented through additional transformations or features. Physiological responses may involve nonlinear interactions, so the model may not capture every pattern present in the data.

In this project, Logistic Regression provides a comparatively simple baseline against which the two tree-based ensemble methods can be evaluated. Its results help establish whether a relatively straightforward decision boundary can distinguish baseline or relaxation observations from stress observations under participant-independent validation.

2.4.2 Random Forest

Random Forest is an ensemble-learning algorithm that combines predictions from multiple decision trees. During training, the trees are constructed using randomized samples and subsets of features. In classification, their predictions are aggregated to produce the final class or class probabilities.

The method can represent nonlinear relationships and interactions among predictors without requiring the user to specify those relationships explicitly. It can also accommodate a mixture of physiological features with different distributions and scales. These properties make Random Forest a useful candidate for structured physiological feature data.

However, ensemble complexity does not guarantee better performance. Random Forest may be sensitive to the amount of available training data, the characteristics of the feature space, and the diversity of observations across participants. When the number of independent participants is small, the effective evidence available for generalization remains limited even if the feature matrix contains many windows or sessions.

In the present study, Random Forest is used to assess whether an ensemble of decision trees improves classification compared with Logistic Regression. Its performance is evaluated using class-sensitive metrics and threshold-independent ranking metrics. This is important because a model may rank observations effectively while producing a different balance of false positives and false negatives at its selected decision threshold.

2.4.3 HistGradientBoosting

HistGradientBoosting is a gradient-boosted decision-tree method that builds an ensemble sequentially. Each new tree is fitted to improve the predictions made by the existing ensemble. The histogram-based approach groups feature values into bins, which can make training more efficient on suitable tabular datasets.

Gradient boosting can learn nonlinear decision boundaries and interactions between physiological features. It may therefore capture relationships that are not adequately represented by a linear classifier. Its flexibility, however, must be evaluated carefully because model performance depends on the data and the settings used during training. A flexible model can still perform poorly when observations from unseen participants differ from those represented in the training data.

HistGradientBoosting is included in this study to evaluate whether sequentially boosted trees provide a performance advantage over the linear and bagged-tree approaches. The comparison is performed independently for WESAD and QoL_Stress, because differences in dataset characteristics may change the relative performance of the algorithms.

The results from this project illustrate why model selection should be empirical rather than based only on algorithmic complexity. HistGradientBoosting performs strongly on the WESAD experiment, whereas Logistic Regression performs best among the three tested models on QoL_Stress. These results are project-specific and do not establish a universal ranking of the algorithms.

2.4.4 Evaluation of Machine Learning Models in Previous Studies

Earlier research demonstrates the feasibility of using wearable physiological measurements to classify experimentally defined stress and affect states. The introduction of WESAD by Schmidt et al. [1] provided a multimodal benchmark that supports the development and comparison of machine-learning approaches. Subsequent research has continued to investigate physiological features, classifier design, and approaches to improving stress detection [6], [7].

However, results reported across studies cannot always be compared directly. Studies may differ in their participants, experimental protocols, sensor devices, signal-processing methods, target definitions, feature sets, class distributions, and validation strategies. A reported accuracy value is therefore meaningful only when interpreted in relation to the experimental setting in which it was obtained.

This issue is particularly important when evaluation uses windows extracted from continuous recordings. If windows from the same participant are randomly divided between training and testing, the model may learn participant-specific patterns that are present in both sets. The resulting score can overstate its ability to classify observations from an entirely new individual. Participant-independent evaluation provides a more demanding test of generalization [4].

Cross-dataset research also highlights the importance of differences in measurement and data distribution. Benchekroun et al. [5] examined cross-dataset generalization for HRV-based stress detection, addressing the question of whether patterns learned from one dataset remain useful in another. Such work motivates evaluating models beyond a single benchmark and being cautious when interpreting strong results from a particular collection of participants or sensors.

The present study follows this comparative perspective by evaluating three classifiers on two distinct datasets. It does not claim to reproduce the performance of every previous method or to establish a universal best classifier. Instead, it uses a consistent evaluation framework to examine how model performance differs between a continuous laboratory-oriented dataset and an episodic consumer-wearable dataset.

2.5 Generalization and Participant-Independent Evaluation

2.5.1 Inter-Participant Physiological Variability

Inter-participant variability refers to differences in physiological measurements and responses among individuals. People may differ in their resting heart rate, HRV patterns, electrodermal responses, and the magnitude of their physiological reactions to a stress-inducing condition. Consequently, a feature pattern associated with stress in one participant may not have the same predictive value for another participant.

This variability creates a challenge for supervised learning. A model may learn relationships that are specific to the individuals represented in its training data instead of identifying patterns that generalize across individuals. The problem becomes particularly important when a dataset contains many observations from a relatively small number of participants.

Vos et al. [4] identify generalization as a central issue in wearable stress monitoring. A model's performance should therefore be assessed under a validation strategy that reflects the intended prediction setting. If the intended setting involves an individual who was not represented during model training, the evaluation should withhold that individual's data from the training process.

The present study addresses this concern by using Leave-One-Subject-Out cross-validation for both WESAD and QoL_Stress. This approach ensures that the participant held out for a particular evaluation fold is not represented in the corresponding training data. It provides a more appropriate estimate of performance on unseen participants than a random observation-level split, although it does not establish performance in every real-world environment.

2.5.2 Participant-Level Data Leakage

Data leakage occurs when information that would not legitimately be available during model training or prediction influences the evaluation process. Leakage can result in performance estimates that are more optimistic than the model's actual ability to generalize.

In physiological stress classification, one important source of leakage is the presence of observations from the same participant in both the training and test sets. Physiological measurements can contain individual-specific characteristics. When windows from a participant appear in both sets, the model may exploit those characteristics instead of learning relationships that transfer to unseen participants.

Overlapping windows create an additional concern. In the WESAD experiment, windows are sixty seconds long and separated by a ten-second stride. Consecutive windows therefore share most of their recorded time interval. If these windows are randomly distributed across training and testing sets, the two sets may contain highly similar signal segments. This can make the evaluation less independent than it appears.

Leakage can also occur through predictor construction. Participant identifiers, target labels, condition strings, timestamps that reveal the experimental phase, or measurements obtained after the target condition may provide information that would not be valid for the intended prediction task. The QoL_Stress preprocessing workflow therefore excludes participant identifiers and the condition field from the predictors and excludes post-condition state-anxiety measurements from the primary physiological feature set.

Preprocessing must also be fitted using training data only. For example, a feature scaler should calculate its parameters from the training portion of each fold and then apply those parameters to the held-out participant. If preprocessing learns from the full dataset before evaluation, information from the held-out data can influence the fitted transformation. Placing preprocessing inside a scikit-learn pipeline helps maintain the separation between training and evaluation data.

These safeguards do not remove every possible source of bias, but they reduce important forms of leakage and improve the credibility of the reported evaluation.

2.5.3 Leave-One-Subject-Out Cross-Validation

Leave-One-Subject-Out cross-validation is a grouped validation method in which one participant is held out for testing during each fold. All observations from the remaining participants are used for training. The procedure is repeated until each participant has served as the held-out participant.

For a dataset containing S participants, the procedure produces Sevaluation folds, assuming every participant is included in the evaluation. The training set for fold s contains observations from every participant except participant s, while the test set contains the observations belonging to participant s.

The method can be represented as:

$$D_{\text{train}}^{(s)} = \{(x_i, y_i): g_i \neq s\}$$

$$D_{\text{test}}^{(s)} = \{(x_i, y_i): g_i = s\}$$

Where x_i represents the feature vector, y_i represents the class label, and g_i identifies the participant associated with observation i.

The key property is that no participant contributes observations to both training and test sets within the same fold. Predictions from the held-out participants can subsequently be combined to calculate pooled out-of-fold metrics.

LOSO is appropriate for the present project because the objective is to examine model performance on participants not used during the corresponding training process. It also supports a consistent evaluation principle across WESAD and QoL_Stress, despite their different observation structures.

There are limitations to this method. When the number of participants is small, the estimate may be sensitive to the characteristics of individual held-out participants. Some participants may also contribute observations from only one class, which can make certain fold-level metrics undefined. In such cases, pooled out-of-fold predictions and fold-level averages should be reported separately rather than treating them as interchangeable quantities.

2.5.4 Cross-Dataset Generalization Challenges

Cross-dataset generalization concerns whether a model or a learned relationship remains useful when the data collection setting changes. For physiological stress detection, datasets may differ in sensor placement, sampling rate, recording duration, preprocessing, participant characteristics, experimental protocols, and class definitions.

WESAD and QoL_Stress illustrate several of these differences. WESAD provides continuous recordings from chest-worn and wrist-worn research-oriented sensors. The project uses a prepared matrix of sixty-second windows with 144 physiological predictors. QoL_Stress provides episodic consumer-wearable measurements represented by 124 session-level observations and 25 primary physiological predictors.

These differences mean that the two experiments do not provide identical inputs to a single common model. Instead, the same three classifier families are trained and evaluated separately within each dataset. This design allows the project to compare their performance under different data conditions without assuming that the feature matrices are directly interchangeable.

The broader literature supports the need for caution when interpreting results across datasets. Benchekroun et al. [5] specifically investigate cross-dataset generalization for HRV-based stress detection. Their research direction is relevant because success within one dataset does not necessarily establish that the same patterns will remain predictive under different collection conditions.

Accordingly, the performance difference observed in this project should be interpreted as a difference between two experimental tracks, not as proof that consumer wearables are inherently incapable of detecting stress. Dataset size, participant composition, feature availability, sensor characteristics, and the experimental protocol may all contribute to the observed results. Establishing true cross-dataset transfer would require a dedicated experiment in which a model is trained on one dataset and evaluated on another using a compatible feature representation and carefully defined labels.

2.6 Evaluation Metrics for Stress Classification

A reliable comparison of stress-classification models requires more than a single accuracy value. In binary classification, the number of correctly classified observations can be misleading when the classes are imbalanced or when the costs of different errors are unequal. A model that frequently predicts the majority class may achieve reasonable accuracy while failing to identify a substantial proportion of stress observations.

For this reason, the present study considers balanced accuracy, precision, recall, macro F1-score, ROC-AUC, and PR-AUC. These metrics measure different aspects of classification performance and provide a more complete basis for comparing the three algorithms.

For the following definitions, stress is treated as the positive class, while baseline or relaxation is treated as the negative class. The exact interpretation of the negative class depends on the dataset being evaluated.

2.6.1 Balanced Accuracy

Balanced accuracy calculates the average of sensitivity and specificity. It gives equal weight to the classification performance of the positive and negative classes, making it useful when the class frequencies differ.

It is defined as:

$$Balanced\ Accuracy = \frac{Sensitivity + Specificity}{2}$$

where:

$$Sensitivity = \frac{TP}{TP+FN}$$

$$Specificity = \frac{TN}{TN+FP}$$

Here, TP represents true positives, TN true negatives, FP false positives, and FN false negatives.

Balanced accuracy is particularly relevant to the WESAD experiment because baseline windows account for 64.89% of the prepared observations, while stress windows account for 35.11%. Although this imbalance is not extreme, a metric that gives equal importance to both classes provides a clearer view of class-specific performance than ordinary accuracy alone.

2.6.2 Precision, Recall, and Macro F1-Score

Precision measures the proportion of observations predicted as stress that are actually labelled as stress. It is calculated as:

$$Precision = \frac{TP}{TP + FP}$$

Recall measures the proportion of actual stress observations correctly identified by the model:

$$Recall = \frac{TP}{TP + FN}$$

These metrics represent different aspects of performance. A model with high precision produces relatively few false-positive predictions among its predicted stress cases. A model with high recall

identifies a relatively large proportion of the stress observations. Improving one metric may come at the expense of the other, depending on the classifier and its decision threshold.

The F1-score combines precision and recall through their harmonic mean:

$$F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

Macro F1-score calculates the F1-score independently for each class and then averages the two class-specific values. This prevents the larger class from dominating the summary metric solely because it contains more observations.

The present study reports macro F1-score alongside balanced accuracy because the two metrics provide complementary information. Balanced accuracy summarizes sensitivity and specificity, while macro F1-score combines precision and recall across both classes. Together, they help identify models that perform unevenly across baseline or relaxation and stress observations.

2.6.3 Receiver Operating Characteristic Area Under the Curve

The receiver operating characteristic area under the curve, commonly called ROC-AUC, measures how well a model ranks positive observations above negative observations across different decision thresholds. It is based on the relationship between the true-positive rate and false-positive rate.

The false-positive rate is defined as:

$$FPR = \frac{FP}{FP + TN}$$

ROC-AUC is useful because it evaluates ranking performance without restricting the assessment to a single classification threshold. A value closer to 1 generally indicates stronger separation between the positive and negative classes, while a value near 0.5 indicates ranking performance close to random ordering.

However, ROC-AUC does not directly specify how many false-positive or false-negative classifications occur at the selected operating threshold. It should therefore be interpreted alongside confusion-matrix-based metrics such as precision, recall, and balanced accuracy.

2.6.4 Precision-Recall Area Under the Curve

The precision-recall area under the curve, or PR-AUC, summarizes the relationship between precision and recall across decision thresholds. It is useful when the positive class is relatively uncommon or when identifying positive cases is a central objective.

Unlike ROC-AUC, PR-AUC directly emphasizes the quality of positive-class predictions. Its interpretation depends on the positive-class prevalence and on the precise implementation used to calculate the area. Values should therefore be compared with an understanding of the dataset's class distribution and should not be treated as universally comparable across datasets.

In the present project, PR-AUC complements ROC-AUC by assessing the trade-off between precision and recall for stress detection. Reporting both measures provides a broader understanding of model ranking performance, while balanced accuracy and macro F1-score describe performance at the evaluated classification threshold.

2.6.5 Precision-Recall Area Under the Curve

A confusion matrix records the numbers of true positives, true negatives, false positives, and false negatives. It makes the types of classification errors visible and supports the calculation of several metrics discussed above.

For stress classification, false negatives correspond to stress observations incorrectly classified as baseline or relaxation. False positives correspond to baseline or relaxation observations incorrectly classified as stress. The relative importance of these errors depends on the intended application and its operating requirements.

Because the present project is an academic predictive-analytics experiment rather than a deployed clinical system, the confusion matrix is used to describe model behaviour, not to define a clinical decision policy. Any future application would require a separate assessment of appropriate thresholds, the consequences of errors, calibration, and human oversight.

2.7 Research Gap

The existing literature establishes that physiological measurements from wearable devices can be used to investigate and classify experimentally labelled stress-related states. WESAD provides a multimodal benchmark for wearable stress and affect detection [1], while research reviews describe the physiological indicators used for stress monitoring and the challenges involved in developing generalizable models [3], [4], [7]. Cross-dataset investigations further highlight that performance may be affected when models encounter data collected under different conditions [5].

Despite this progress, several methodological issues remain important. First, high performance on a controlled dataset does not automatically establish that a model will perform similarly on consumer-wearable data. Differences in sensor configuration, recording duration, feature availability, and experimental protocol can affect the patterns available for classification. Second, the number of windows or sessions should not be confused with the number of independent participants. Participant-level overlap can lead to overly optimistic performance estimates if validation is not designed appropriately. Third, model evaluation based on a single metric may obscure differences between false-positive and false-negative errors or uneven performance across classes.

A further challenge is that datasets with different observation structures cannot always be compared directly. Continuous recordings can support window-based feature extraction, whereas short or episodic recordings require session-level analysis. Differences in sample size and participant composition also affect the uncertainty of the evaluation. Consequently, observed performance differences should be interpreted in the context of the complete experimental design rather than attributed to one factor without further testing.

The research gap addressed by the present project is therefore methodological and comparative. The project evaluates three machine-learning classifiers, Logistic Regression, Random Forest, and HistGradientBoosting, on two public datasets with different sensing and recording characteristics. It applies participant-independent LOSO validation, places preprocessing within the training pipeline, excludes identified leakage variables, and reports multiple complementary evaluation metrics.

The project does not claim to introduce a new classification algorithm or to establish clinical validity. Its contribution is to provide a structured comparison of the implemented models under participant-aware evaluation and to examine how their performance differs between the WESAD and QoL_Stress experimental settings. A separate synthetic-data experiment is also conducted, but its results are kept distinct from the human-data experiments and are not used as evidence of real-world stress detection.

The following research questions guide this comparison:

1. How do Logistic Regression, Random Forest, and HistGradientBoosting compare in classifying baseline or relaxation and stress observations under participant-independent validation?
2. How does model performance differ between the continuous WESAD window-based experiment and the episodic QoL_Stress session-based experiment?
3. Why is participant-aware validation important when physiological observations are collected repeatedly from the same individuals?
4. How do the selected metrics describe differences in class-wise classification performance?
5. What is the effect of adding demographic and baseline trait-stress features to the primary physiological feature set in the QoL_Stress ablation experiment?
6. How should the synthetic-data results be interpreted separately from results obtained using human physiological data?

These questions connect the published research to the specific experiments implemented in this project. They also define the limits of the conclusions that can be drawn from the available data.

2.8 Summary of the Literature Review

The reviewed literature demonstrates the potential of wearable physiological signals for studying experimentally defined stress-related states. ECG, HRV, and EDA provide complementary measurements, but their relationship with stress is influenced by individual variability, recording conditions, and the experimental protocol. WESAD offers a multimodal laboratory-oriented benchmark, while QoL_Stress provides a different setting based on episodic consumer-wearable measurements [1], [2].

The literature also highlights the importance of generalization and evaluation design. Participant overlap between training and testing data can lead to overly optimistic results, particularly when repeated or overlapping observations are used. Participant-independent validation, leakage prevention, and multiple complementary metrics are therefore necessary for a more credible assessment of model performance [4], [5].

Based on these considerations, the present study compares Logistic Regression, Random Forest, and HistGradientBoosting separately on WESAD and QoL_Stress using LOSO cross-validation. Balanced accuracy, macro F1-score, precision, recall, ROC-AUC, PR-AUC, and confusion matrices are used to examine different aspects of classification performance. The synthetic experiment is treated as a separate supplementary analysis.

The next chapter presents the workflow and methodology, including dataset preparation, feature selection, preprocessing, model training, participant-independent validation, and evaluation procedures used to conduct the experiments.

CHAPTER 3: WORKFLOW AND METHODOLOGY

3.1 Overview of the Proposed Workflow

his project follows a structured predictive analytics workflow to investigate the classification of experimentally labelled stress-related states using physiological measurements. The workflow includes dataset selection, data quality assessment, preprocessing, feature preparation, exploratory data analysis, model training, participant-independent validation, and performance evaluation. Two real-world datasets with different recording characteristics are analysed independently to examine how the selected machine-learning models perform under different experimental conditions.

The first experimental track uses the Wearable Stress and Affect Detection (WESAD) dataset, which contains continuous physiological recordings collected using research-oriented wearable sensors. The prepared dataset consists of 2,575 sixty-second windows from 15 participants, represented by 144 physiological predictors. The classification task distinguishes baseline observations from stress observations, with the amusement class excluded from this binary experiment.

The second experimental track uses the QoL_Stress dataset, which contains episodic physiological measurements collected using a consumer fitness tracker. The prepared feature matrix consists of 124 session-level observations from 64 participants and 25 primary physiological predictors derived from ECG and electrodermal activity. The classification task distinguishes relaxation sessions from stress sessions. Because the datasets differ in their recording protocols, observation units, and available features, each dataset is processed and modelled separately.

Three supervised machine-learning algorithms are evaluated in both experimental tracks: Logistic Regression, Random Forest, and HistGradientBoosting. These models provide a comparison between a linear classification approach and two tree-based ensemble approaches. Their performance is assessed using Leave-One-Subject-Out (LOSO) cross-validation, in which observations from one participant are held out for testing while the remaining participants are used for training.

Preprocessing is incorporated into the model pipeline so that transformations requiring fitted parameters are learned from the training data within each fold. Identifiers and variables that could reveal the target are excluded from the predictor set. Evaluation uses multiple metrics, including balanced accuracy, macro F1-score, precision, recall, ROC-AUC, and PR-AUC, to assess different aspects of classification performance.

In addition to the two experiments using human physiological data, a separate synthetic-data experiment is conducted. This experiment evaluates the selected classifiers on simulated observations and is reported independently. Its results are not combined with the WESAD or QoL_Stress results and are not treated as evidence of real-world human stress detection.

The overall workflow is organized into the following stages:

1. Dataset selection and examination of the available physiological measurements.
2. Data quality assessment, cleaning, and preparation of dataset-specific feature matrices.
3. Exploratory analysis of the observations, predictors, and target-class distributions.
4. Definition of the binary classification tasks and exclusion of inappropriate predictor variables.
5. Training of Logistic Regression, Random Forest, and HistGradientBoosting models.

6. Participant-independent evaluation using LOSO cross-validation.
7. Calculation and comparison of classification metrics and error patterns.
8. Separate evaluation of the synthetic dataset and interpretation of the limitations of the experiments.

This workflow is designed to provide a consistent and reproducible basis for comparing the selected models while respecting the differences between the datasets. The detailed procedures used at each stage are described in the following sections.

3.2 Dataset Selection and Description

Two publicly available datasets, WESAD and QoL_Stress, were selected for the empirical analysis. Both contain physiological measurements associated with experimentally defined stress-related conditions, but they differ in sensor configuration, recording duration, observation structure, and feature availability. Using both datasets provides an opportunity to evaluate the selected classifiers under two distinct experimental settings.

The datasets are analysed independently rather than merged into a single feature matrix. Their physiological predictors are not directly interchangeable because the sensors, measurement characteristics, and feature-extraction procedures differ. The study therefore compares the performance of the same model families across separate experiments instead of assuming that the datasets contain equivalent observations.

3.2.1 WESAD Dataset

WESAD, or Wearable Stress and Affect Detection, is a multimodal dataset introduced by Schmidt et al. [1]. It contains physiological recordings collected from 15 participants using chest-worn and wrist-worn sensors during controlled baseline, stress, and amusement conditions.

The project uses a prepared feature matrix derived from the physiological recordings. The signals are divided into sixty-second windows with a ten-second stride, and statistical features are extracted from the selected chest and wrist channels. Windows that do not satisfy the required label-purity condition are excluded from the binary classification dataset.

The resulting dataset contains 2,575 observations and 144 physiological predictors. Each observation represents one sixty-second window. The target variable distinguishes baseline from stress, with baseline encoded as class 0 and stress encoded as class 1. The amusement condition is excluded from this experiment.

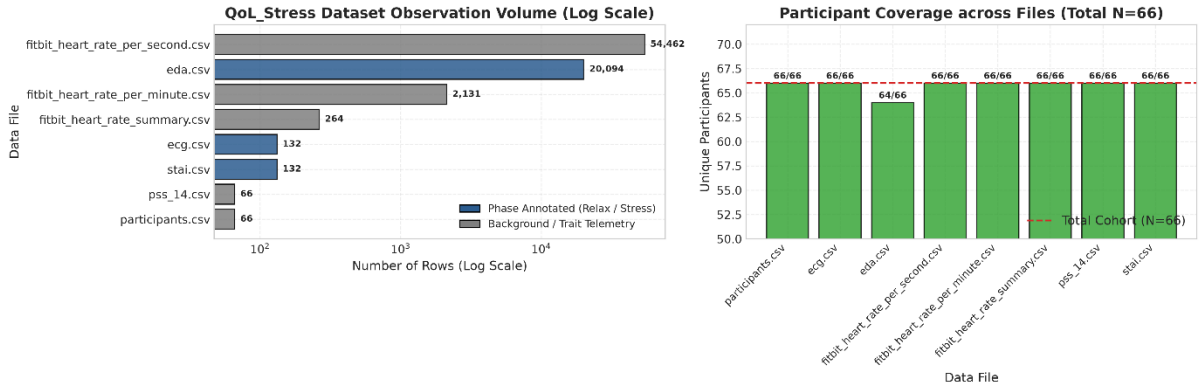
3.2.2 QoL_Stress Dataset

QoL_Stress is a multimodal dataset available through PhysioNet and described by Calvo et al. [2]. It includes physiological and self-reported measurements associated with controlled relaxation and acute-stress conditions. The project uses measurements collected with the Fitbit Charge 6 consumer fitness tracker.

Unlike WESAD, QoL_Stress is represented at the session level rather than through overlapping sliding windows. The prepared feature matrix contains 124 observations from 64 unique participants. The primary predictor set comprises 25 physiological features: 15 ECG and HRV features and 10 EDA features.

The target variable distinguishes relaxation, encoded as class 0, from stress, encoded as class 1. Participant identifiers, condition metadata, and post-condition state-anxiety scores are excluded

from the primary predictor matrix. Demographic and baseline trait-stress variables are reserved for a separate feature-ablation experiment.



3.2.3 Dataset Comparison

Table 3.1 summarizes the principal characteristics of the two datasets as used in this project.

Table 3.1: Comparison of the datasets used in the empirical experiments.

Characteristic	**WESAD**	**QoL_Stress**
Source	Schmidt et al. [1]	Calvo et al. [2]
Sensor setting	Research-oriented chest and wrist sensors	Consumer fitness tracker
Observation unit	Sixty-second window	Experimental session
Window or recording structure	Continuous recordings, 10 second stride	Episodic ECG and EDA recordings
Number of observations	2,575	124
Number of participants	15	64
Primary physiological predictors	144	25
Target classes	Baseline and stress	Relaxation and stress
Target encoding	0 = Baseline, 1 = Stress	0 = Relaxation, 1 = Stress
Validation method	LOSO cross-validation	LOSO cross-validation

The comparison highlights an important methodological distinction. WESAD provides a larger number of observations because continuous recordings are converted into overlapping windows. QoL_Stress contains fewer observations because the analysis uses session-level measurements. The 2,575 WESAD windows should not be interpreted as 2,575 independent participants, just as the 124 QoL_Stress sessions should not be interpreted as 124 independent individuals.

The differences in dataset size, feature representation, and measurement conditions are considered when interpreting the results. The experiments are designed to compare model performance within each dataset and examine the differences between the two experimental settings. They do not establish that the observed performance difference is caused by sensor type alone.

3.3 Data Loading and Quality Assessment

Data loading and quality assessment were performed before model training to establish the structure, completeness, and suitability of the prepared feature matrices. This stage involved examining the available observations, checking the predictor and target columns, identifying participant metadata, and assessing potential data-quality problems.

For WESAD, the preprocessing workflow reads the participant recordings, examines the available sensor signals and labels, and constructs the window-based feature matrix. For QoL_Stress, the workflow examines the available ECG, EDA, and related session information before producing the session-level feature matrix.

The quality-assessment stage is important because errors in parsing, missing values, incorrect label alignment, or unsuitable predictors can affect the reliability of subsequent modelling. In particular, physiological recordings must be associated with the correct experimental labels before supervised learning is performed.

3.3.1 WESAD Data Quality Assessment

The WESAD workflow processes participant recordings sequentially and extracts features from windows that satisfy the selected duration and label-purity requirements. The prepared feature matrix was checked for missing and infinite values, and the final dataset contained no missing values, infinite values, or duplicate observations according to the recorded quality audit.

The target distribution was also examined to determine the balance between baseline and stress windows. This check is important because class imbalance can influence the interpretation of model performance and motivates the use of balanced accuracy and macro F1-score alongside other metrics.

3.3.2 QoL_Stress Data Quality Assessment

The QoL_Stress workflow examines session-level ECG and EDA measurements and prepares features aligned with the experimental conditions. The raw ECG recordings required parsing checks because some waveform values were represented using tokenized strings containing isolated hyphen delimiters. The preprocessing workflow reconstructed the affected values before feature extraction.

The workflow also assessed EDA recording quality and the availability of measurements across sessions. Background heart-rate telemetry without reliable experimental phase timestamps was excluded from supervised target mapping because assigning condition labels heuristically could introduce label ambiguity.

The resulting feature matrix was checked for missing and infinite values, duplicate observations, target-label consistency, and participant coverage. The prepared matrix contained 124 session-level observations with no missing or infinite values and no duplicate observations according to the recorded audit.

3.3.3 Quality Checks and Their Purpose

Table 3.2 summarizes the principal data-quality checks applied during dataset preparation.

Table 3.2: Data-quality checks performed before model evaluation.

Quality check	**Purpose**

Observation and column counts	Confirm the structure of the prepared feature matrix
Missing-value inspection	Identify incomplete predictor values
Infinite-value inspection	Identify invalid numerical values
Duplicate-observation check	Detect repeated rows in the prepared matrices
Target-label verification	Confirm that observations are mapped to the intended binary classes
Participant-identifier inspection	Preserve the grouping information required for LOSO
Predictor audit	Exclude identifiers, target-derived variables, and unsuitable metadata
Recording and feature checks	Identify dataset-specific parsing, alignment, or sensor-quality concerns

These checks support the integrity of the prepared data but do not guarantee that every possible source of measurement error has been eliminated. The limitations associated with participant coverage, controlled experimental conditions, and differences in sensor characteristics remain relevant to the interpretation of the results.

3.4 Data Preprocessing and Feature Preparation

Data preprocessing and feature preparation convert raw physiological recordings into structured numerical observations suitable for supervised machine learning. The procedures differ between WESAD and QoL_Stress because one dataset is represented through continuous sliding windows, while the other is represented through individual experimental sessions.

The preprocessing stage focuses on preserving the relationship between physiological measurements and their experimental labels, constructing numerical predictors, and excluding variables that could compromise model evaluation. The resulting feature matrices are used as inputs to the three classifiers described in Section 3.7.

3.4.1 WESAD Windowing and Feature Extraction

The WESAD recordings contain continuous physiological signals collected from chest-worn and wrist-worn sensors. To represent these recordings in a tabular format, the signals were divided into fixed-duration windows, and statistical features were calculated for each selected channel.

A window duration of 60 seconds and a stride of 10 seconds were used. Since each new window begins 10 seconds after the previous window starts, adjacent windows overlap substantially. The overlap proportion is calculated as:

$$Overlap = \left(1 - \frac{Stride}{Window\ duration}\right) \times 100$$

Substituting the selected values:

$$Overlap = \left(1 - \frac{10}{60}\right) \times 100 = 83.33\%$$

The overlapping-window approach increases the number of observations available for analysis while retaining a consistent temporal duration for feature extraction. However, adjacent windows are not independent because they share much of the same underlying signal. This is one reason participant-independent validation is used in the modelling stage.

The feature extraction procedure calculates nine statistical descriptors for each of 16 selected chest and wrist channels. These descriptors include mean, standard deviation, minimum, maximum, median, 25th percentile, 75th percentile, interquartile range, and root mean square. The resulting feature matrix contains 144 physiological predictors.

Only windows satisfying the specified label-purity condition were retained. Windows containing transitions between target conditions or observations belonging to excluded affective states were not included in the final binary classification dataset. Each retained window was associated with its participant identifier, window index, temporal boundaries, and target label. These metadata fields support tracking and validation but are not used as physiological predictors.

The final prepared WESAD matrix contains 2,575 observations. Each observation represents a sixty-second window labelled as baseline or stress. The extracted features provide a standardized tabular representation of the selected physiological signals for the subsequent machine-learning experiments.

3.4.2 QoL_Stress Session-Level Feature Preparation

The QoL_Stress dataset uses a different preparation strategy because its measurements are collected in short, episodic recordings rather than continuous streams suitable for the same sliding-window procedure.

ECG waveform recordings and EDA sessions were processed to construct session-level physiological features. The ECG processing workflow included reconstruction of malformed waveform tokens where necessary, followed by extraction of waveform and cardiac-related features. The EDA workflow produced features describing skin-conductance measurements and their variability.

The primary feature matrix contains 25 physiological predictors, comprising 15 ECG and HRV features and 10 EDA features. These predictors represent cardiac activity, beat-to-beat variation, waveform characteristics, and electrodermal responses. Each row corresponds to a labelled relaxation or stress session.

Background heart-rate records without reliable timestamps connecting them to the experimental conditions were not used to create supervised session labels. This restriction prevents uncertain temporal associations from being treated as reliable ground truth.

The final prepared dataset contains 124 observations from 64 participants. Unlike the WESAD matrix, which contains overlapping windows, this dataset contains session-level observations. Therefore, the two feature matrices are evaluated independently rather than combined directly.

3.4.3 Feature Selection and Leakage Prevention

Feature selection establishes which variables are permitted to enter the model. The target label and tracking information must be separated from the predictor matrix so that the model learns from the intended physiological measurements rather than from information that reveals the answer.

For WESAD, the participant identifier, window index, start time, end time, and target label are maintained as metadata or target information rather than included among the 144 physiological predictors.

For QoL_Stress, the primary predictor matrix contains the 25 physiological features. The participant identifier and experimental condition field are excluded from the predictors, and the target label is used only as the outcome to be predicted. The post-condition state-anxiety score is also excluded from the primary physiological experiment because it is measured after the condition and could introduce target-related information.

Demographic variables and the baseline Perceived Stress Scale score are reserved for a separate feature-ablation experiment. This design makes it possible to assess the effect of adding these covariates without mixing them into the primary physiological-only comparison.

Participant identifiers are retained separately because they are required to define the LOSO groups. Excluding them from the feature matrix does not mean discarding them from the evaluation procedure.

3.4.4 Missing Values, Scaling, and Pipeline Preparation

Data quality checks were performed on the prepared feature matrices before model training. The recorded quality audits found no missing values or infinite values in either final matrix. Consequently, the final prepared datasets did not require missing-value imputation for the reported experiments.

Feature scaling was handled within the scikit-learn modelling pipeline. This is important because physiological features can have different numerical ranges, and scaling may affect the optimization of algorithms such as Logistic Regression.

For each LOSO fold, the preprocessing transformation is fitted using the training participants and then applied to the held-out participant. The held-out observations are not used to estimate the transformation parameters. This prevents information from the evaluation participant from influencing the fitted preprocessing stage.

The tree-based models do not generally require feature scaling in the same way as linear models, but the study maintains a consistent pipeline-based evaluation design where appropriate to the implemented model configuration. All model comparisons use the prepared dataset-specific predictors and the same participant-grouping principle.

The completed preprocessing stage produces two separate feature matrices ready for model training: a window-based WESAD matrix and a session-based QoL_Stress matrix. Their different observation structures are preserved throughout the evaluation process.

3.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the structure, distribution, and characteristics of the physiological data before model training. The analysis focused on class distribution, participant-level observations, physiological signal patterns, feature distributions, and relationships among predictors. Separate analyses were conducted for the WESAD and QoL_Stress datasets because they differ in their recording protocols, sensor characteristics, and data structures.

The purpose of EDA was to identify patterns relevant to stress classification, understand potential data quality issues, and examine differences between baseline or relaxation conditions and stress conditions. These observations also helped establish the limitations of comparing results across the two datasets.

3.5.1 WESAD Exploratory Analysis

The WESAD dataset was examined at the participant level and at the window level after preprocessing and feature extraction. The final binary classification dataset contained 2,575 windows, each represented by 144 physiological features. The target variable consisted of two classes: baseline and stress. The amusement class was excluded from the final binary classification task.

The distribution of the target classes was examined to determine whether the dataset was balanced. Of the 2,575 windows, 1,671 belonged to the baseline class and 904 belonged to the stress class. Therefore, baseline accounted for 64.89% of the observations, while stress accounted for 35.11%.

Table 3.4 presents the target class distribution.

Table 3.4: Target Class Distribution in the WESAD Dataset

Class	**Number of windows**	**Percentage**
Baseline	1,671	64.89%
Stress	904	35.11%
Total	2,575	100.00%

The class distribution shows that the baseline class contains more observations than the stress class. Although the dataset is not severely imbalanced, the difference is relevant during model evaluation. Accuracy alone may not fully describe performance when the classes have unequal representation. Therefore, balanced accuracy, macro F1-score, precision, and recall were included in the evaluation framework.

The participant-level distribution was also examined to understand how the two classes were represented across the participants. This analysis is important because physiological responses may vary between individuals. A model could perform well on observations from participants whose patterns are represented in training but perform less consistently on an unseen participant.

The participant-wise analysis provides context for the participant-independent evaluation used later in the workflow. In particular, it helps distinguish overall class representation from the distribution of observations belonging to individual participants. Participant identifiers were retained for grouping during validation but were excluded from the model predictors.

The physiological signal traces were also compared between baseline and stress conditions. This analysis was used to visually inspect whether the recorded physiological signals exhibited differences between the two conditions. Such differences may provide useful information for classification, although the visual appearance of a signal alone does not establish that a model can reliably distinguish stress in unseen individuals.

The signal distribution analysis provided a complementary view by comparing the distributions of selected physiological measurements between the two classes. Comparing distributions can help identify features whose values vary across conditions and can reveal overlapping patterns between classes. Overlap is particularly important because physiological responses are influenced by individual characteristics and experimental conditions.

A feature correlation matrix was also examined to understand relationships among the extracted predictors. Since nine statistical features were calculated for each of the 16 physiological channels, some features may represent related aspects of the same underlying signal. Correlation analysis helps identify such relationships and provides a general understanding of the feature space.

The correlation matrix was used for exploratory interpretation rather than as the sole basis for removing predictors. Correlated features do not automatically need to be removed for tree-based models, and the final feature selection process also considered target leakage, participant identifiers, and the requirements of the validation procedure.

3.5.2 QoL_Stress Exploratory Analysis

EDA was performed separately on the QoL_Stress dataset to examine its cohort composition, condition distribution, physiological measurements, and questionnaire variables. Unlike WESAD, which was transformed into overlapping time windows, QoL_Stress was represented using session-level observations for the primary modelling experiment.

The final modelling dataset contained 124 session rows from 64 unique participants. The two conditions were represented by 64 relaxation sessions and 60 stress sessions. Sixty participants had paired relaxation and stress observations, while four participants had relaxation-only observations. Table 3.5 summarises the condition distribution.

Table 3.5: Condition Distribution in the QoL_Stress Dataset

Condition	**Number of sessions**	**Percentage**
Relaxation	64	51.61%
Stress	60	48.39%
Total	124	100.00%

The condition distribution is relatively balanced, with a small difference in the number of relaxation and stress observations. However, the number of sessions should not be confused with the number of independent participants. Several observations belong to the same participant, making participant-level grouping necessary during model validation.

The cohort characteristics and availability of observations were also examined. This analysis provided an overview of the participant composition and helped identify the distinction between participant-level information and session-level measurements. Demographic variables were not included in the primary physiological feature set; they were reserved for a secondary feature-ablation experiment.

Questionnaire scores were explored to understand the psychological measurements available in the dataset. These variables provide contextual information about perceived stress and anxiety. However, psychological questionnaire variables do not necessarily measure the same phenomenon as the physiological features recorded during a particular session.

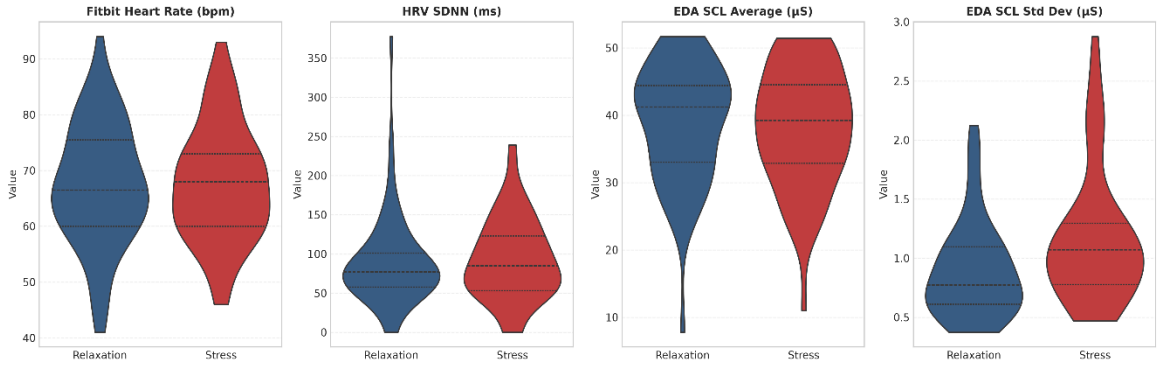
The notebook-based analysis also examined state-anxiety scores in relation to the experimental conditions. The recorded analysis found higher state-anxiety scores under the stress condition, with a Wilcoxon test result of $W = 1481.5$ and $p = 3.82 \times 10^{-4}$.This observation supports the distinction between the experimental conditions at the questionnaire level. However, it does not establish the predictive performance of the physiological classification models.

State-anxiety measurements collected after a condition were excluded from the primary model predictors because they could reveal information about the experimental condition being predicted. Including such variables could create target leakage and produce misleading evaluation results.

The physiological signals were also inspected across relaxation and stress sessions. The dataset included episodic Lead-I ECG recordings and electrodermal activity measurements. The ECG

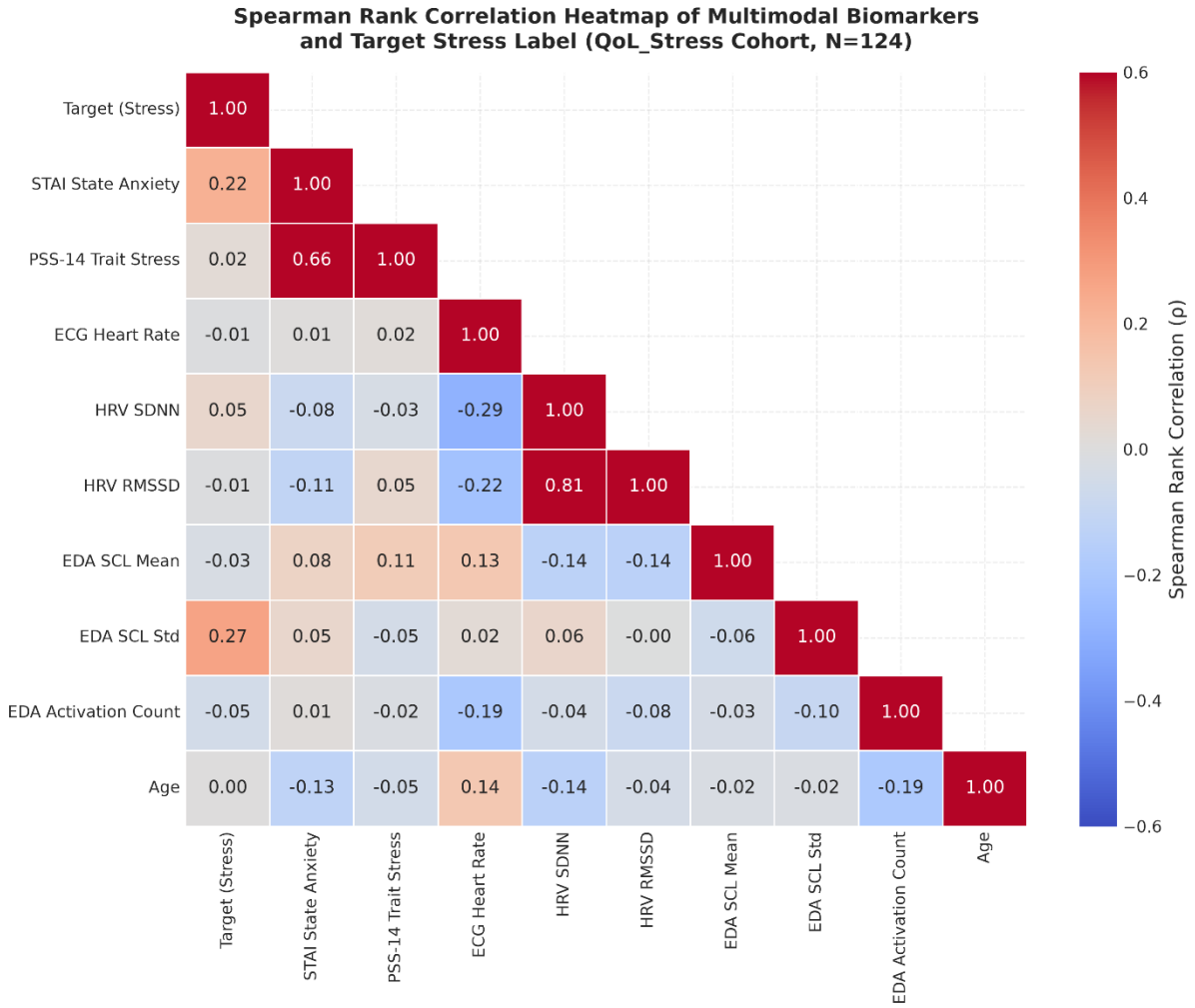
recordings were sampled at 250 Hz, while the recorded EDA sessions were approximately 2.7 minutes long at 1 Hz, according to the project's dataset preparation analysis.

The distributions of selected physiological features were compared between conditions to examine potential differences in cardiovascular and electrodermal measurements. The analysis included ECG and heart rate variability features as well as EDA-derived features. These comparisons were exploratory and were not treated as proof that any single feature could independently identify stress.



The exploratory analysis also examined electrodermal activity variability. The notebook-based statistical analysis reported a significant increase in the standard deviation of skin conductance level during the stress condition, with $p = 0.0023$ and an effect size of $r = 0.271$. This finding describes an observed difference in the analysed sessions. It should not be interpreted as evidence that all physiological features or all participants respond identically to stress.

A feature correlation matrix was examined to understand relationships among the physiological predictors.



The correlation analysis helped describe the relationships among ECG, HRV, and EDA-derived features. Given the relatively small number of session observations, model evaluation was designed to preserve participant independence rather than relying only on random splitting of individual rows.

3.5.3 Interpretation of Dataset Differences

The EDA results highlight important differences between the two datasets. WESAD provides a larger window-level dataset with multiple physiological channels and a comparatively larger number of observations. QoL_Stress provides a smaller session-level dataset based on measurements collected under controlled relaxation and acute stress conditions.

Table 3.6 summarises the principal differences relevant to the modelling workflow.

Table 3.6: Comparison of EDA Findings for WESAD and QoL_Stress

Aspect	**WESAD**	**QoL_Stress**
Unit of analysis	60-second windows	Session-level observations
Final observations	2,575 windows	124 sessions
Unique participants	15	64
Target conditions	Baseline and stress	Relaxation and stress

Class distribution	Baseline: 64.89%; stress: 35.11%	Relaxation: 51.61%; stress: 48.39%
Primary physiological predictors	144 statistical features	25 physiological features
Main EDA focus	Class distribution, participant variation, signal patterns, feature relationships	Cohort composition, questionnaire scores, physiological patterns, feature relationships
Important modelling consideration	Overlapping windows and participant dependence	Small session count and repeated observations from participants

The differences in recording protocol and data representation mean that the two datasets cannot be treated as interchangeable. WESAD's overlapping windows increase the number of observations available for modelling, but neighbouring windows may share substantial portions of the original signals. QoL_Stress contains fewer observations and uses session-level measurements, which limits the amount of data available for estimating model performance.

The datasets also differ in their sensor characteristics, experimental protocols, and class definitions. Consequently, performance values obtained from one dataset should not be directly compared with those from the other without considering these differences.

The EDA findings informed three important methodological decisions. First, participant identifiers were retained for validation grouping but excluded from the predictors. Second, the target class was defined consistently within each dataset without combining the original class labels into a single cross-dataset target. Third, model performance was evaluated using participant-independent validation and multiple metrics rather than accuracy alone.

Overall, EDA established the structure and class distribution of both datasets, provided a descriptive understanding of their physiological measurements, and helped identify considerations that needed to be addressed before training and evaluation.

3.6 Target Definition and Feature Selection

Target definition and feature selection were important steps in preparing the datasets for supervised machine learning. The target variable determines the condition that the model must predict, while the predictor variables provide the physiological information used to make that prediction. Since WESAD and QoL_Stress differ in their experimental protocols and data structures, target labels and feature sets were defined separately for each dataset.

For WESAD, the classification task was defined as distinguishing baseline from stress. For QoL_Stress, the task was to distinguish relaxation from stress. The two datasets were modelled independently, and their observations were not combined into a single training dataset.

In WESAD, the amusement condition was excluded because the project focused on binary stress classification. The remaining observations were assigned binary target labels representing baseline and stress. The final modelling dataset contained 2,575 windows, with 1,671 baseline observations and 904 stress observations.

For QoL_Stress, relaxation and stress sessions were assigned the corresponding target labels. The final dataset contained 124 session-level observations, comprising 64 relaxation sessions and 60 stress sessions. The primary modelling experiment used 25 physiological predictors derived from ECG, heart rate variability, and electrodermal activity measurements. Table 3.7 summarises the target definitions and primary feature sets used in the project.

Table 3.7: Target Definition and Primary Feature Selection

Component	**WESAD**	**QoL_Stress**
Prediction task	Baseline versus stress classification	Relaxation versus stress classification
Negative class	Baseline	Relaxation
Positive class	Stress	Stress
Final observations	2,575 windows	124 sessions
Primary predictors	144 statistical features	25 physiological features
Feature construction	Nine statistical measures calculated across 16 physiological channels	Fifteen ECG/HRV features and ten EDA features
Participant identifier	Retained for validation grouping, excluded from predictors	Retained for validation grouping, excluded from predictors
Additional variables	Window metadata excluded from predictors	Demographic and questionnaire variables reserved for secondary analysis

The feature selection process also addressed potential target leakage. Variables that directly identified the participant, encoded the target condition, or represented window timing were not used as model predictors. In WESAD, participant identifiers, window indices, start and end times, and target labels were excluded from the predictor matrix. Participant identifiers were retained separately to define the groups required for Leave-One-Subject-Out validation.

For QoL_Stress, participant identifiers and condition-related metadata were excluded from the physiological predictor set. State-anxiety measurements obtained after the experimental condition were also excluded from the primary predictors because they could provide information about the condition being classified. Including such information could lead to an unrealistically favourable evaluation.

Demographic and psychological variables, including age, sex, and PSS-14 score, were reserved for a secondary feature-ablation experiment. This experiment examined whether adding these variables changed model performance compared with the primary physiological feature set. These additional variables were therefore evaluated separately rather than being included in the primary experiment.

Table 3.8 summarizes the principal inclusion and exclusion decisions.

Table 3.8: Feature Inclusion and Exclusion Rules

Variable category	**WESAD**	**QoL_Stress**

Physiological features	Included in primary model	Included in primary model
Participant identifier	Excluded from predictors; used for grouping	Excluded from predictors; used for grouping
Target label	Used as the prediction target only	Used as the prediction target only
Window or condition metadata	Excluded from predictors	Excluded from predictors
Post-condition state anxiety measurements	Not applicable to the defined feature set	Excluded to prevent leakage
Demographic and psychological variables	Not part of the primary feature set	Reserved for secondary ablation

The final feature sets were designed to ensure that predictions were based on the intended physiological measurements rather than identifiers or variables that could reveal the answer. This distinction was particularly important because the project evaluated generalisation to participants not used during model training.

Overall, target definition and feature selection established consistent binary classification tasks within each dataset and reduced the risk of misleading evaluation results. The resulting predictor sets were then used for model development and participant-independent validation.

3.7 Machine Learning Models

Three supervised machine learning algorithms were implemented to classify physiological observations into stress and non-stress conditions: Logistic Regression, Random Forest, and HistGradientBoosting. These models were selected to compare a linear classification approach with two tree-based ensemble approaches.

The models were evaluated separately on the WESAD and QoL_Stress datasets. The same overall evaluation framework was used to examine their ability to distinguish the target conditions, while accounting for differences in dataset size, feature representation, and participant-level variability.

The implementation used scikit-learn pipelines where appropriate to maintain a consistent preprocessing and modelling workflow. Feature scaling was fitted using training data within each validation fold to avoid information from the held-out participant influencing model preparation. The models were evaluated using Leave-One-Subject-Out (LOSO) validation, described in Section 3.8.

3.7.1 Logistic Regression

Logistic Regression was used as the baseline classification algorithm. It is a supervised learning method that estimates the probability of an observation belonging to a particular class. In this project, it estimated the probability that a physiological observation represented stress rather than baseline or relaxation.

For a binary classification problem, Logistic Regression applies the sigmoid function to a linear combination of the input features. Its mathematical representation is given by Equation (3.1).

$$P(y = 1 \mid \boldsymbol{x}) = \frac{1}{1+exp\left[-\left(\beta_0+\sum_{j=1}^{p}\beta_j x_j\right)\right]}$$

where:

- $P(y = 1 \mid \mathbf{x})$ is the estimated probability of the stress class.
- x_j represents the value of the j-th predictor.
- β_j is the coefficient associated with that predictor.
- β_0 is the intercept.
- $p$is the total number of predictors.

The predicted class is determined by comparing the estimated probability with the classification threshold used by the implementation. Logistic Regression provides a useful baseline because it models a linear relationship between the predictors and the log-odds of the positive class. Its performance can therefore be compared with that of more flexible tree-based algorithms.

The model was trained using the selected physiological features from each dataset. For WESAD, the input consisted of 144 statistical features extracted from physiological signals. For QoL_Stress, the primary model used 25 physiological features derived from ECG, heart rate variability, and EDA measurements.

Feature scaling was incorporated into the preprocessing pipeline where required by the implemented model. During LOSO evaluation, the pipeline was fitted using only the training participants in each fold. This prevented the held-out participant's observations from contributing to the fitted scaling parameters.

Logistic Regression was included because it offers a relatively simple reference point for evaluating the predictive information contained in the physiological features. Its results also provide a basis for examining whether the more complex ensemble models deliver improvements under participant-independent validation.

3.7.2 Random Forest

Random Forest is an ensemble learning algorithm that combines the predictions of multiple decision trees to produce a final classification. Each tree is trained using a sample of the training data, while candidate features are considered during tree construction. The individual predictions are combined to obtain the final output. Random Forest is based on the principle of aggregating multiple trees to improve predictive performance and reduce dependence on a single tree [10].

For binary classification, the predicted class can be obtained by aggregating the class predictions of the individual trees. The probability of the positive class can also be estimated from the proportion of trees predicting that class, depending on the implementation.

The classification process can be represented as:

$$\hat{y} = \text{mode}\{T_1(\mathbf{x}), T_2(\mathbf{x}), \dots, T_B(\mathbf{x})\}$$

where:

- $\hat{y}$ is the predicted class.
- $T_b(\mathbf{x})$ is the prediction produced by the b-th decision tree.
- B is the total number of trees.
- mode represents the majority class among the tree predictions.

Random Forest was used to examine whether combining multiple decision trees could capture relationships in the physiological features that might not be adequately represented by a linear model. The algorithm can model nonlinear relationships and interactions between predictors without requiring the same linear decision boundary assumed by Logistic Regression.

In this project, Random Forest was trained independently on the WESAD and QoL_Stress feature sets. The WESAD model used 144 statistical predictors, whereas the primary QoL_Stress model used 25 physiological predictors. Participant identifiers were excluded from the predictor matrices and retained separately for LOSO validation.

Random Forest was evaluated using the same validation strategy and performance metrics as Logistic Regression. This made it possible to compare the two approaches under the same participant-independent evaluation framework.

3.7.3 HistGradientBoosting

HistGradientBoosting is a gradient boosting classification algorithm that builds an ensemble of decision trees sequentially. Instead of constructing independent trees and combining them through voting, gradient boosting builds successive trees to improve the predictions of the existing ensemble. HistGradientBoosting uses histogram-based processing of feature values, which can make tree construction more computationally efficient for suitable datasets [9], [11].

The general gradient boosting process can be expressed as an additive model:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta h_m(\mathbf{x})$$

where:

- $F_m(\mathbf{x})$ is the ensemble prediction after adding the m-th tree.
- $F_{m-1}(\mathbf{x})$ is the prediction from the previous ensemble.
- $h_m(\mathbf{x})$ is the contribution of the newly added tree.
- η is the learning rate, which controls the contribution of each new tree.

This equation describes the general additive structure of gradient boosting. The actual training procedure also depends on the loss function and the implementation's configuration.

HistGradientBoosting was included to investigate whether sequentially improving tree-based predictions could provide better stress classification performance than Logistic Regression and Random Forest. It can represent nonlinear patterns in the data and interactions among physiological features.

The model was trained separately on the primary feature sets of WESAD and QoL_Stress. The same target definitions and participant-level validation procedure were maintained so that the results could be compared with those of the other algorithms.

The model's performance was assessed using balanced accuracy, macro F1-score, precision, recall, ROC-AUC, PR-AUC, and confusion matrices where applicable. These metrics provide complementary information about classification quality, particularly when the classes differ in size or when missing stress cases is an important concern.

3.7.4 Summary of the Selected Models

Table 3.9 summarises the three machine learning algorithms used in the experiments.

Table 3.9: Summary of Machine Learning Models

Model	Learning approach	Main characteristics	Purpose in this project
Logistic Regression	Linear probabilistic classifier	Simple baseline, interpretable coefficients, requires an appropriate linear decision boundary in the transformed feature space	Establish baseline classification performance
Random Forest	Bagging-based tree ensemble	Combines multiple trees and can represent nonlinear relationships	Evaluate ensemble learning using independent tree predictions
HistGradientBoosting	Sequential tree boosting	Builds trees successively and uses histogram-based processing	Evaluate an alternative ensemble approach that improves predictions iteratively

All three models were evaluated using the same dataset-specific target definitions and LOSO validation strategy. This helped ensure that differences in their measured performance were not caused by using different participant splits.

The models were not assumed to perform equally well on both datasets. Differences in sample size, sensor measurements, experimental conditions, and participant variability can affect their performance. The results are therefore reported separately for WESAD and QoL_Stress in Chapter 4.

3.8 Participant-Independent Validation Using LOSO

The evaluation strategy is an important component of physiological stress classification because physiological patterns can differ substantially between individuals. A model evaluated using randomly divided windows or sessions may encounter observations from the same participant in both the training and testing sets. This can produce overly optimistic estimates of generalisation performance.

To address this issue, the project used Leave-One-Subject-Out (LOSO) cross-validation. In this approach, all eligible observations belonging to one participant are held out for testing, while observations from the remaining participants are used for training. The procedure is repeated until every participant has been held out.

For a dataset containing N unique participants, the LOSO procedure can be represented as:

$$D = \bigcup_{i=1}^{N} D_i$$

where D_i represents the observations associated with participant i. During fold i, the training and testing sets are defined as:

$$D_{\text{train}}^{(i)} = D \setminus D_i$$

$$D_{\text{test}}^{(i)} = D_i$$

Here, $D_{\text{train}}^{(i)}$ contains observations from all participants except participant i, while $D_{\text{test}}^{(i)}$ contains observations from the held-out participant.

This validation strategy was applied separately to the WESAD and QoL_Stress datasets. For WESAD, the grouping variable was the participant identifier associated with each extracted window. For QoL_Stress, the participant identifier was used to group session-level observations. Identifiers were used for constructing the folds but were excluded from the model predictors.

The implementation also ensured that preprocessing operations requiring fitted parameters were performed within the training fold. In particular, feature scaling was fitted using the training participants and then applied to the held-out participant. This prevented information from the test participant from influencing the fitted scaling parameters.

For WESAD, the extracted windows used a duration of 60 seconds and a stride of 10 seconds. Consequently, adjacent windows could overlap substantially. Randomly splitting these windows could place highly similar segments from the same participant in both training and testing sets. LOSO reduces this participant-level leakage risk by ensuring that all windows belonging to the held-out participant remain outside the training set.

For QoL_Stress, multiple sessions could belong to the same participant. Grouping by participant ensured that paired relaxation and stress observations from the same individual were not divided between the training and testing sets.

Table 3.10 summarises the validation strategy used in the experiments.

Table 3.10: LOSO Validation Design

Component	**WESAD**	**QoL_Stress**
Final observations	2,575 windows	124 sessions
Unique participants	15	64
Grouping variable	Participant identifier	Participant identifier
Held-out unit	One participant and all associated windows	One participant and all associated sessions
Training data	Observations from remaining participants	Observations from remaining participants
Preprocessing	Fitted within the training fold where applicable	Fitted within the training fold where applicable
Prediction coverage	100% pooled out-of-fold coverage	100% pooled out-of-fold coverage
Main objective	Estimate performance on an unseen participant	Estimate performance on an unseen participant

After the fold-level predictions were generated, the held-out predictions were combined to calculate pooled out-of-fold metrics. This allowed every eligible observation to contribute a prediction from a model that had not been trained on that observation's participant.

An important consideration for QoL_Stress is that some held-out participants had observations from only one condition. Consequently, certain fold-level metrics that require both classes could be undefined. The project therefore distinguishes pooled out-of-fold metrics from averages of fold-level metrics. These quantities should not be treated as interchangeable.

LOSO provides a more stringent estimate of participant-independent performance than a random observation-level split. However, it does not guarantee generalisation to every population, device, or real-world setting. The results remain dependent on the characteristics of the available participants, experimental protocols, and physiological measurements.

3.9 Evaluation Metrics

The three classification models were evaluated using multiple metrics to capture different aspects of predictive performance. Accuracy alone was not considered sufficient because the datasets differed in class distribution and the practical importance of correctly identifying stress may differ from that of correctly identifying the non-stress condition.

The evaluation included balanced accuracy, macro F1-score, precision, recall, ROC-AUC, PR-AUC, and confusion matrices. These metrics were calculated from the held-out predictions generated through LOSO validation. Pooled out-of-fold metrics were used to summarise predictions across the eligible observations.

3.9.1 Balanced Accuracy and Macro F1-Score

Balanced accuracy calculates the average recall across the two classes. It gives equal weight to the ability to identify positive and negative cases, making it useful when the class distribution is unequal.

For binary classification, balanced accuracy is defined as:

$$Balanced\ Accuracy = \frac{Sensitivity + Specificity}{2}$$

where sensitivity is the proportion of actual stress cases correctly identified and specificity is the proportion of actual non-stress cases correctly identified.

Macro F1-score calculates the F1-score separately for each class and then takes their arithmetic mean. This gives both classes equal weight in the final metric.

$$F1_{macro} = \frac{F1_{stress} + F1_{non\text{-}stress}}{2}$$

These metrics were used to compare how consistently the models classified both conditions rather than allowing performance on the more frequent class to dominate the evaluation.

3.9.2 Precision & Recall

Precision measures the proportion of observations predicted as stress that actually belong to the stress class. Recall measures the proportion of actual stress observations that the model correctly identifies.

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

where:

- TP represents true positives.
- FP represents false positives.
- FN represents false negatives.

Precision is relevant because a model that produces many false stress alerts may be less useful in practice. Recall is relevant because missed stress observations may limit the model's ability to identify the target condition. These measures must be interpreted together because improving one can sometimes reduce the other.

3.9.3 ROC-AUC and PR-AUC

The Area Under the Receiver Operating Characteristic Curve (ROC-AUC) measures how well a model ranks positive cases above negative cases across different classification thresholds. The ROC curve plots the true positive rate against the false positive rate.

The false positive rate is defined as:

$$FPR = \frac{FP}{FP + TN}$$

where TN represents true negatives.

The Area Under the Precision-Recall Curve (PR-AUC) summarises the relationship between precision and recall across different thresholds. It is useful for examining positive-class detection, particularly when the classes are imbalanced.

Both ROC-AUC and PR-AUC were used to complement the threshold-dependent classification metrics. Their interpretation depends on the evaluation data and class distribution. In this project, the scores were calculated using pooled out-of-fold predictions rather than by treating training predictions as test results.

3.9.4 Confusion Matrix

A confusion matrix summarises the numbers of correctly and incorrectly classified observations. For the binary stress classification task, it contains four outcomes.

Table 3.11: Structure of the Binary Classification Confusion Matrix

Actual class	**Predicted non-stress**	**Predicted stress**

Non-stress	True Negative (TN)	False Positive (FP)
Stress	False Negative (FN)	True Positive (TP)

True negatives and true positives represent correct predictions. False positives are non-stress observations incorrectly classified as stress, while false negatives are stress observations incorrectly classified as non-stress.

The confusion matrix helps explain model behaviour beyond a single performance score. For example, two models with similar balanced accuracy may differ in the number of missed stress cases and false stress alerts. Examining these differences supports a more meaningful comparison of the classification approaches.

The confusion matrices and numerical metrics are presented in Chapter 4, where the results of each dataset and model are analysed separately.

3.10 Synthetic-Data Experiment

In addition to the experiments conducted using the WESAD and QoL_Stress datasets, a separate synthetic-data experiment was performed to examine the behaviour of the selected machine learning models on simulated physiological observations. This experiment provided a controlled setting for evaluating the classification pipeline independently of the two human datasets. The synthetic dataset contained 2,500 simulated observations, 50 synthetic subjects, and 10 input features. The observations were divided equally between the two target classes, with 1,250 observations assigned to each class. The dataset was generated for experimental evaluation and did not represent actual measurements collected from human participants. Table 3.12 summarises the synthetic dataset used in the experiment.

Table 3.12: Summary of the Synthetic Dataset

Characteristic	**Description**
Total observations	2,500
Synthetic subjects	50
Input features	10
Class 0 observations	1,250
Class 1 observations	1,250
Class distribution	50% per class
Validation approach	Leave-One-Subject-Out
Purpose	Separate evaluation of the classification workflow on simulated data

The same three classification algorithms were evaluated: Logistic Regression, Random Forest, and HistGradientBoosting. Leave-One-Subject-Out validation was used to maintain subject-level

separation between the training and testing observations. Performance was assessed using balanced accuracy, macro F1-score, ROC-AUC, PR-AUC, and confusion matrices.

The synthetic-data experiment was treated as a separate evaluation rather than being combined with the WESAD or QoL_Stress observations. This separation was important because simulated data may have different statistical properties from real physiological measurements. Performance on synthetic data therefore cannot be interpreted as direct evidence of performance on human subjects.

The results of the synthetic experiment are summarised in Table 3.13.

Table 3.13: Performance Metrics for the Synthetic-Data Experiment

Model	**Balanced accuracy**	**Macro F1-score**	**ROC-AUC**	**PR-AUC**
Logistic Regression	0.8964	0.8964	0.9583	0.9625
Random Forest	0.8884	0.8884	0.9504	0.9532
HistGradientBoosting	0.8836	0.8836	0.9471	0.9548

Logistic Regression achieved the highest balanced accuracy and macro F1-score among the three models in this experiment. It also achieved the highest ROC-AUC and PR-AUC. Random Forest and HistGradientBoosting produced slightly lower scores on these metrics.

These results describe the behaviour of the models on the particular simulated dataset and should not be generalised to real-world stress detection. The experiment was included to provide an additional evaluation of the modelling workflow, not to replace the experiments performed on human physiological data.

The synthetic-data experiment complements the main experiments by providing a separate reference point for the classification pipeline. The primary conclusions about physiological stress classification are based on the results from WESAD and QoL_Stress, which are analysed independently in Chapter 4.

3.11 Implementation Environment and Reproducibility

The project was implemented in Python using a notebook-based workflow. Separate notebooks were used for dataset preparation, exploratory analysis, model training, evaluation, and the synthetic-data experiment. This organisation allowed the processing and modelling stages to be examined independently while maintaining a consistent overall workflow.

The implementation was organised into four notebooks:

1. WESAD data selection, exploratory analysis, and preprocessing.
2. QoL_Stress data selection, exploratory analysis, and preprocessing.
3. Machine learning model training, evaluation, comparison, and feature-ablation analysis.
4. Synthetic-data generation and evaluation.

The implementation used Python-based data analysis and machine learning tools, including pandas, NumPy, Matplotlib, and scikit-learn. These tools supported tabular data preparation, numerical computation, visualisation, classification, and evaluation. Scikit-learn provided the modelling and

evaluation framework, including the pipeline and cross-validation components used in the experiments [9].

Table 3.14 summarises the principal implementation components.

Table 3.14: Implementation Environment and Tools

Component	Tool or approach	Purpose
Programming language	Python	Data processing and model implementation
Notebook environment	Jupyter notebooks	Organised execution of project stages
Data manipulation	pandas, NumPy	Data preparation and numerical operations
Visualisation	Matplotlib	Exploratory analysis and result visualisation
Machine learning	scikit-learn	Model training, pipelines, and evaluation
Classification algorithms	Logistic Regression, Random Forest, HistGradientBoosting	Binary stress classification
Validation	Leave-One-Subject-Out	Participant-independent evaluation
Performance analysis	Balanced accuracy, macro F1-score, precision, recall, ROC-AUC, PR-AUC, confusion matrices	Model comparison

Reproducibility was supported by separating data preparation from model evaluation and by maintaining consistent target definitions within each dataset. The two human datasets were processed and modelled independently, and the synthetic experiment was kept separate from the empirical experiments.

The validation procedure was designed to prevent participant-level overlap between the training and held-out sets. Participant identifiers were retained for grouping but excluded from the model predictors. Where feature scaling was required, the scaling parameters were fitted using only the training portion of each fold.

The project also excluded variables that could reveal the target condition directly. In QoL_Stress, post-condition state-anxiety measurements were excluded from the primary physiological feature set. The secondary feature-ablation experiment was used to examine the effect of adding demographic and questionnaire variables separately.

Table 3.15 summarises the main reproducibility and evaluation safeguards.

Table 3.15: Reproducibility and Leakage-Prevention Measures

Measure	Implementation in the project
Dataset separation	WESAD, QoL_Stress, and synthetic data evaluated separately

Participant-level validation	LOSO used to hold out one participant at a time
Predictor preparation	Identifiers and target-derived information excluded from model inputs
Feature scaling	Fitted within the training fold where applicable
Out-of-fold evaluation	Held-out predictions combined for pooled evaluation
Secondary feature analysis	Demographic and psychological variables evaluated separately for QoL_Stress
Synthetic-data interpretation	Simulated results not combined with human-dataset results

The use of pooled out-of-fold predictions also supported consistent reporting across the experiments. However, some fold-level metrics for QoL_Stress could not be calculated when a held-out participant had observations from only one class. These cases were handled by distinguishing pooled metrics from fold-level summaries rather than assuming every metric was defined in every fold.

For full reproducibility, the project should retain the executed notebooks, dataset preparation specifications, configuration files, dependency information, and generated evaluation outputs alongside the final report. The report should describe only those settings and procedures that were actually implemented. Any additional information, such as exact software versions or model hyperparameters, should be recorded from the project environment rather than estimated.

3.12 Chapter Summary

This chapter described the workflow used to develop and evaluate machine learning models for physiological stress classification. The experiments used two human datasets, WESAD and QoL_Stress, alongside a separate synthetic-data experiment.

The WESAD dataset was transformed into 2,575 windows represented by 144 statistical features extracted from 16 physiological channels. The target classes were baseline and stress. The QoL_Stress dataset contained 124 session-level observations from 64 participants, with relaxation and stress as the target conditions. Its primary modelling experiment used 25 physiological predictors derived from ECG, heart rate variability, and electrodermal activity measurements.

Exploratory analysis was conducted to examine class distributions, participant representation, physiological signal patterns, questionnaire measurements, and relationships among features. The resulting feature sets were prepared to exclude identifiers, metadata, and target-derived information that could lead to data leakage.

Three supervised classification algorithms were implemented: Logistic Regression, Random Forest, and HistGradientBoosting. Their performance was assessed using multiple metrics, including balanced accuracy, macro F1-score, precision, recall, ROC-AUC, PR-AUC, and confusion matrices.

Participant-independent evaluation was performed using Leave-One-Subject-Out validation. This approach ensured that observations from the held-out participant were not used to train the corresponding model. Preprocessing operations requiring fitted parameters were also restricted to the training fold where applicable.

Finally, the synthetic-data experiment provided a separate evaluation of the classification workflow using simulated observations. Its results were kept separate from the human-dataset experiments to avoid drawing unsupported conclusions about real-world physiological stress detection.

The methodology established in this chapter provides the basis for the experimental results and model comparisons presented in Chapter 4.

CHAPTER 4: RESULT & DISCUSSION

4.1 Introduction

This chapter presents the results of the machine learning experiments conducted using the WESAD dataset, QoL_Stress dataset, and a separate synthetic dataset. Logistic Regression, Random Forest, and HistGradientBoosting were evaluated using Leave-One-Subject-Out (LOSO) validation.

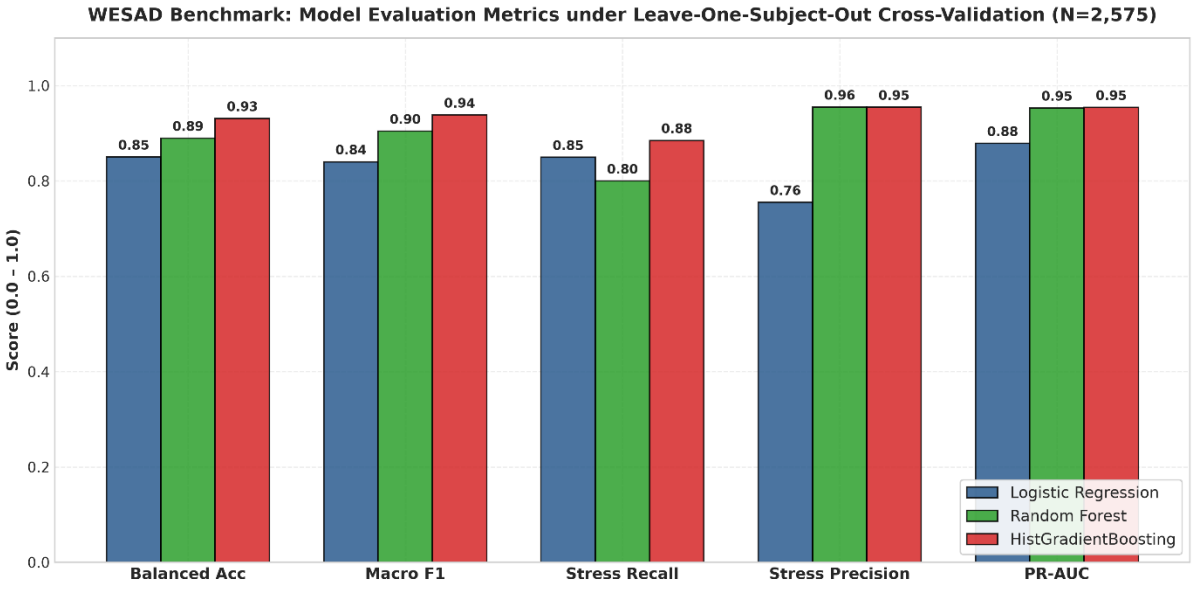
The models were compared using balanced accuracy, macro F1-score, precision, recall, ROC-AUC, PR-AUC, and confusion matrices. The results are presented separately for each dataset because their sample sizes, physiological measurements, and experimental protocols differ.

4.2 Result on WESAD Dataset

The WESAD experiment used 2,575 windows represented by 144 physiological features. The target classes were baseline and stress. HistGradientBoosting achieved the highest balanced accuracy and macro F1-score, while Random Forest achieved the highest ROC-AUC. The confusion matrices also show differences in the models' ability to identify stress observations.

Table 4.1: Performance Comparison on WESAD

Model	**Balanced accuracy**	**Macro F1-score**	**Stress recall**	**ROC-AUC**	**PR-AUC**
Logistic Regression	0.8503	0.8402	0.8496	0.9138	0.8787
Random Forest	0.8897	0.9045	0.7998	0.9739	0.9531
HistGradientBoosting	0.9311	0.9384	0.8850	0.9646	0.9544

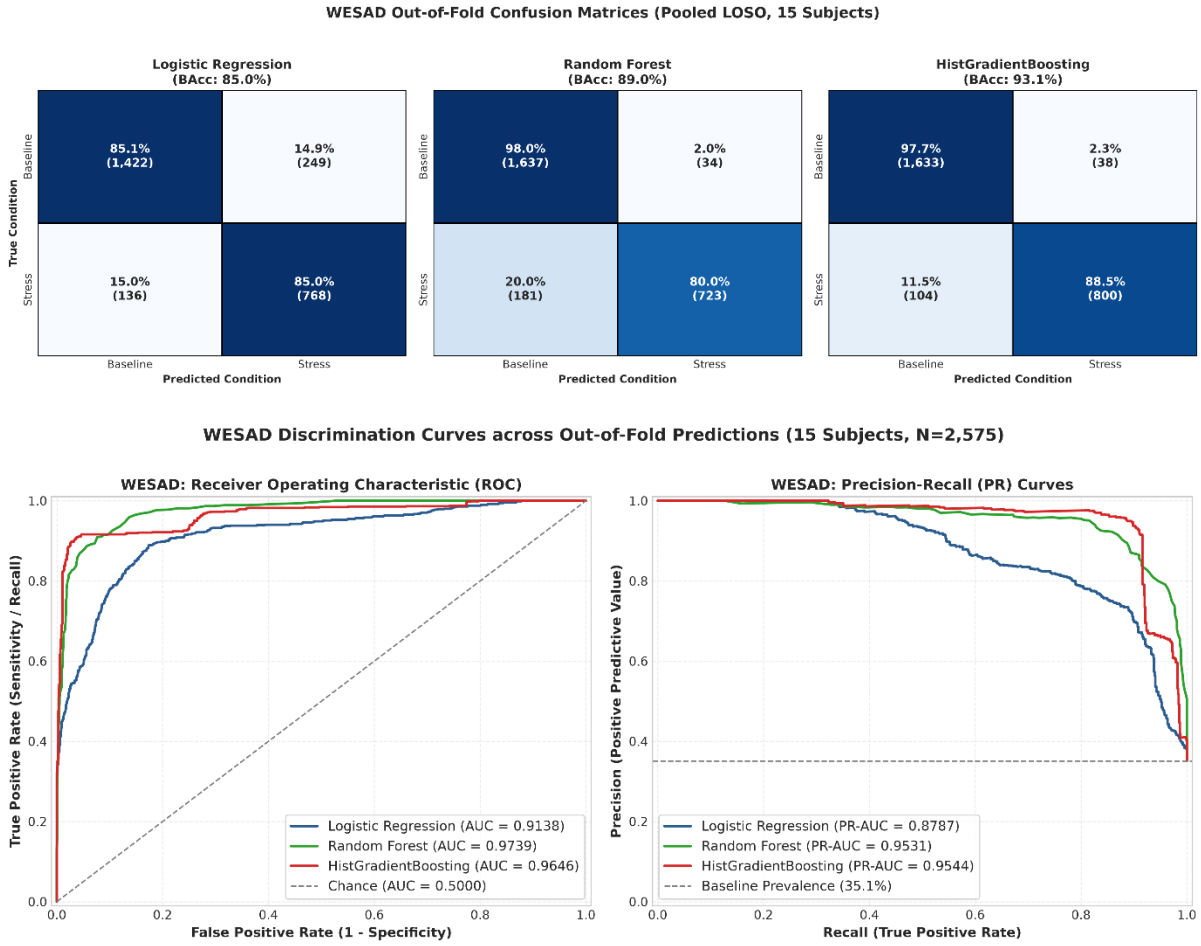


HistGradientBoosting achieved the highest balanced accuracy of 0.9311 and macro F1-score of 0.9384. It also achieved a stress recall of 0.8850, indicating that it correctly identified a high proportion of actual stress windows. Random Forest achieved the highest ROC-AUC of 0.9739, slightly exceeding the HistGradientBoosting score.

These results indicate that HistGradientBoosting performed best on WESAD for balanced accuracy and macro F1-score, whereas Random Forest performed best in ROC-AUC. The preferred model therefore depends on the performance measure being prioritised.

Table 4.2: WESAD Confusion Matrices

Model	TN	FP	FN	TP
Logistic Regression	1,422	249	136	768
Random Forest	1,637	34	181	723
HistGradientBoosting	1,633	38	104	800



Random Forest produced the fewest false positives, with 34, but missed more stress windows than HistGradientBoosting. HistGradientBoosting identified 800 stress windows correctly and had 104 false negatives. This illustrates the trade-off between false alerts and missed stress observations.

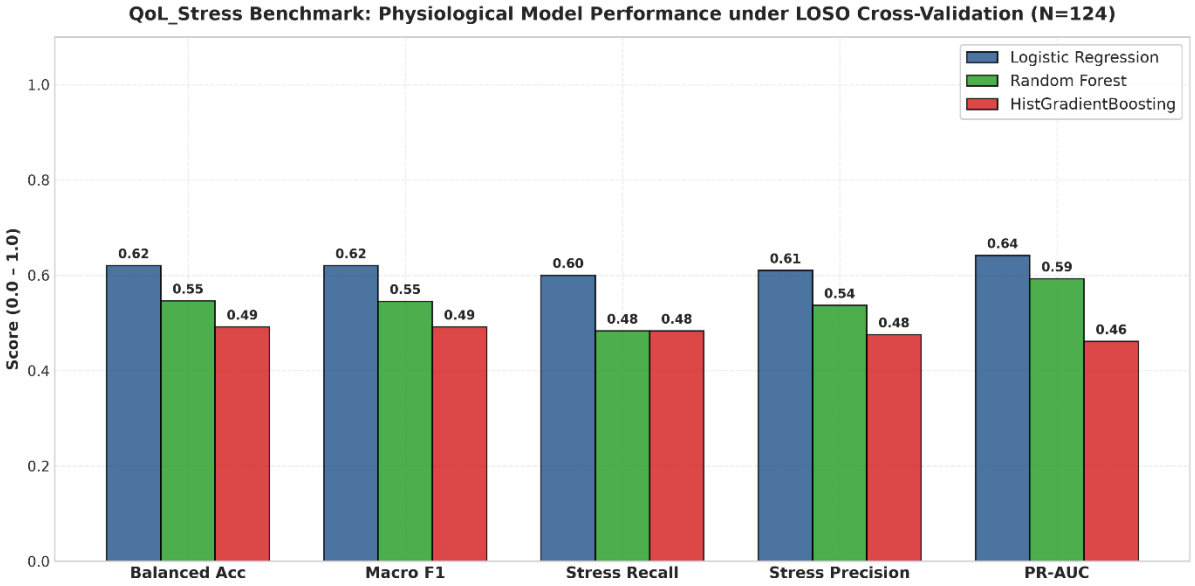
4.3 Result on QoL_Stress Dataset

The QoL_Stress experiment used 124 session-level observations from 64 participants, represented by 25 primary physiological features. Logistic Regression achieved the highest balanced accuracy, macro F1-score, stress recall, ROC-AUC, and PR-AUC among the three models.

Table 4.3: Performance Comparison on QoL_Stress

Model	Balanced accuracy	Macro F1-score	Stress recall	ROC-AUC	PR-AUC

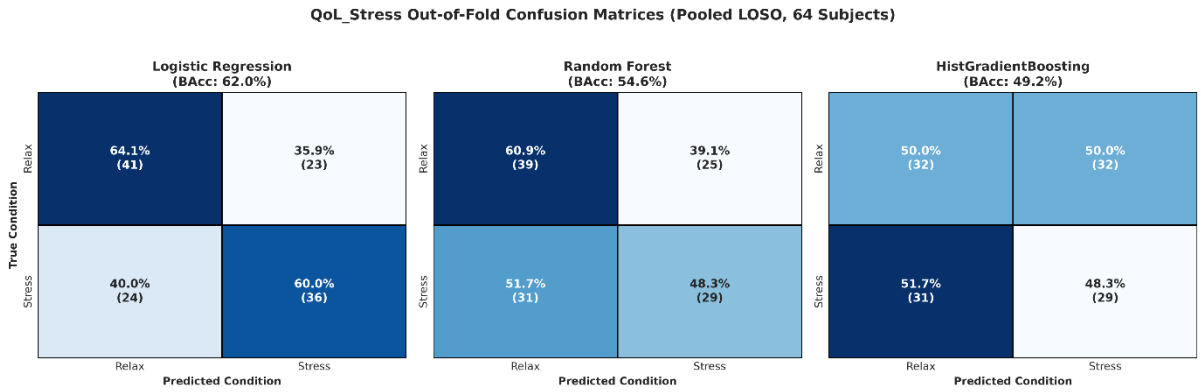
Logistic Regression	0.6203	0.6204	0.6000	0.6625	0.6416
Random Forest	0.5464	0.5454	0.4833	0.6204	0.5926
HistGradientBoosting	0.4917	0.4916	0.4833	0.4958	0.4612



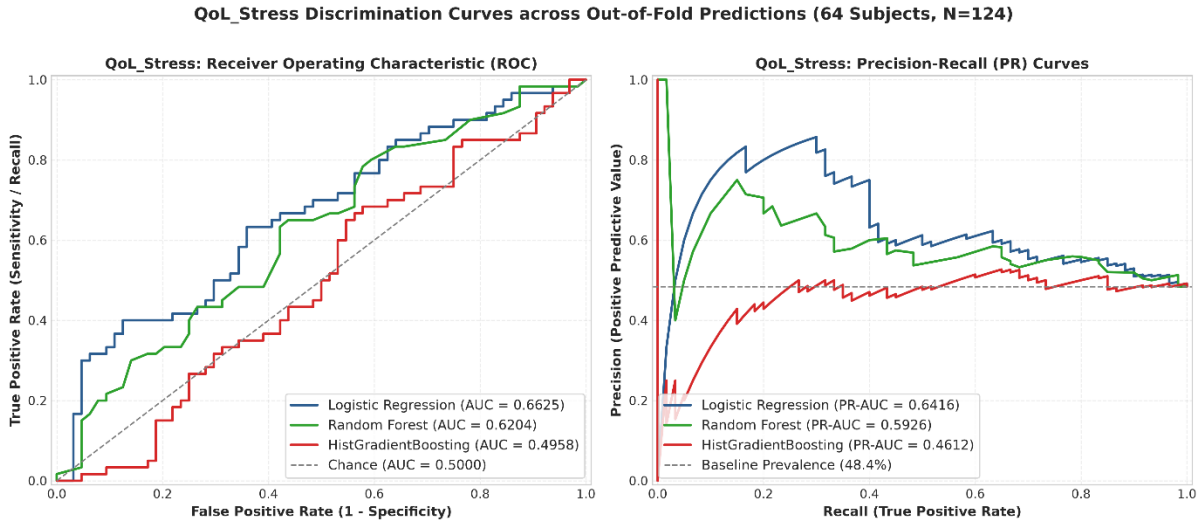
Logistic Regression performed better than the two tree-based models on this dataset. However, its balanced accuracy of 0.6203 indicates that classification remained challenging. HistGradientBoosting achieved a ROC-AUC of 0.4958, which was close to chance-level ranking performance.

Table 4.4: QoL_Stress Confusion Matrices

Model	**TN**	**FP**	**FN**	**TP**
Logistic Regression	41	23	24	36
Random Forest	39	25	31	29
HistGradientBoosting	32	32	31	29



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Logistic Regression correctly identified 36 stress sessions and 41 relaxation sessions. The results indicate that the physiological features provided some useful predictive information, but the models did not achieve the stronger performance observed on WESAD.

The smaller number of session-level observations, differences in recording protocols, and participant variability may contribute to the difficulty of this classification task. These results should be interpreted within the limits of the dataset and its controlled experimental conditions [2].

4.4 Result on Synthetic Dataset

The synthetic-data experiment contained 2,500 simulated observations, 50 synthetic subjects, and 10 input features. The results were evaluated separately from the human datasets.

Table 4.5: Performance Comparison on Synthetic Data

Model	**Balanced accuracy**	**Macro F1-score**	**ROC-AUC**	**PR-AUC**
Logistic Regression	0.8964	0.8964	0.9583	0.9625
Random Forest	0.8884	0.8884	0.9504	0.9532
HistGradientBoosting	0.8836	0.8836	0.9471	0.9548

Figure 03: Synthetic Model Performance Comparison Across Architectures (50 Folds LOSO)

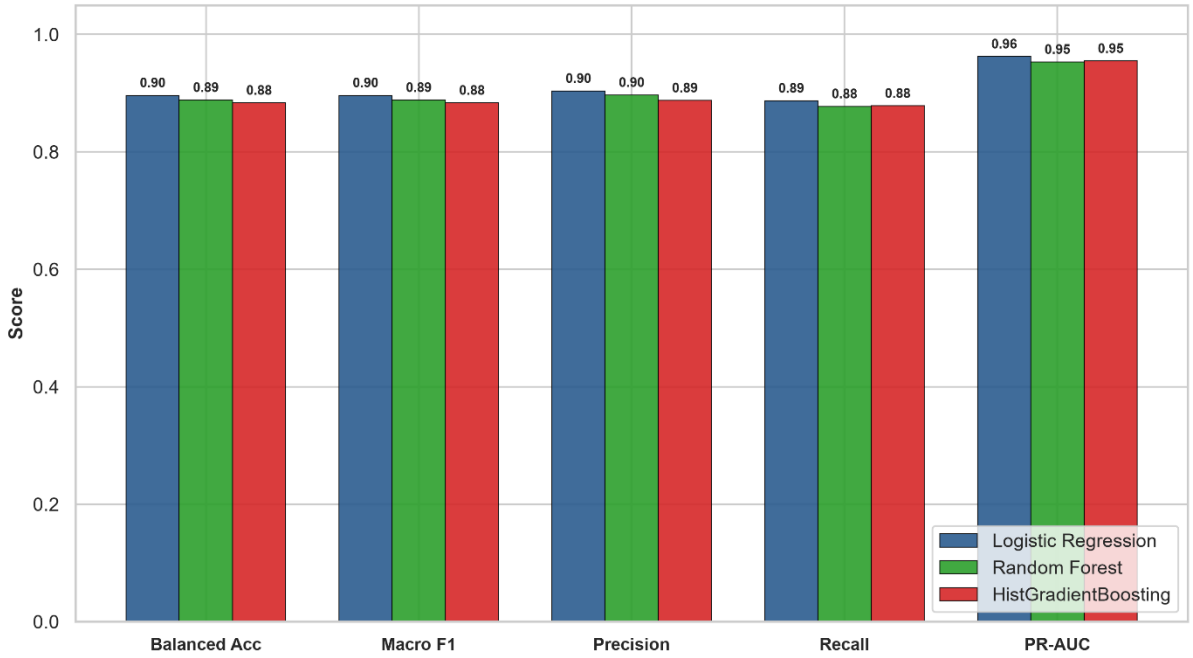


Figure 04: Synthetic Out-of-Fold Confusion Matrices Across Models (N = 2,500)

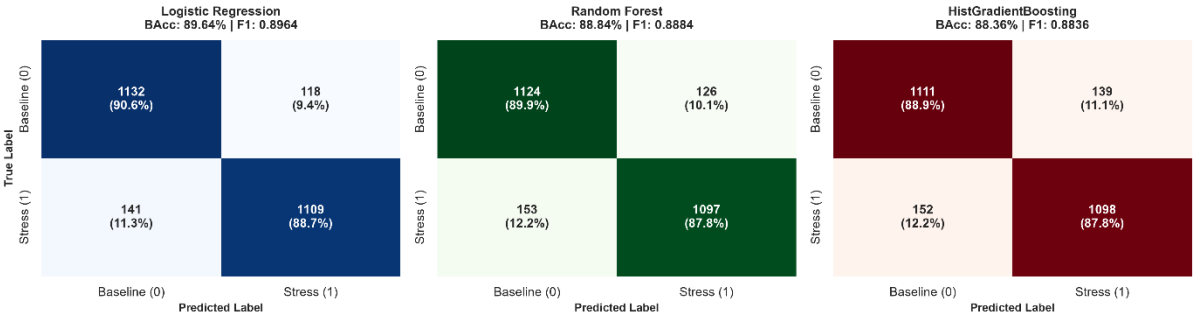
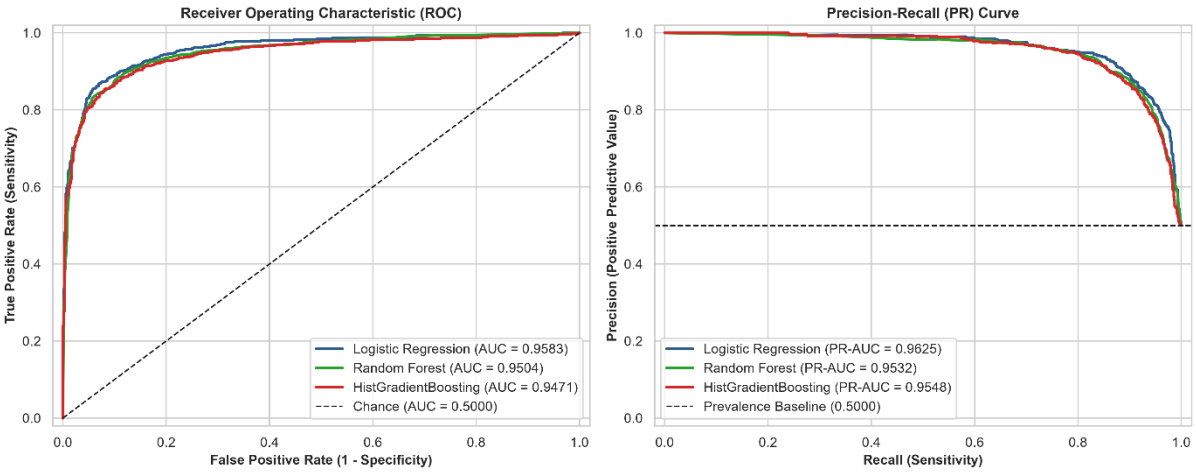


Figure 05: Synthetic Out-of-Fold Discrimination Curves (50 Folds LOSO)



Logistic Regression achieved the highest scores across the four reported metrics. Random Forest and HistGradientBoosting also achieved strong classification results on the simulated data.

These findings show that the models could classify the generated observations effectively under the synthetic experiment's conditions. However, synthetic observations do not reproduce all the

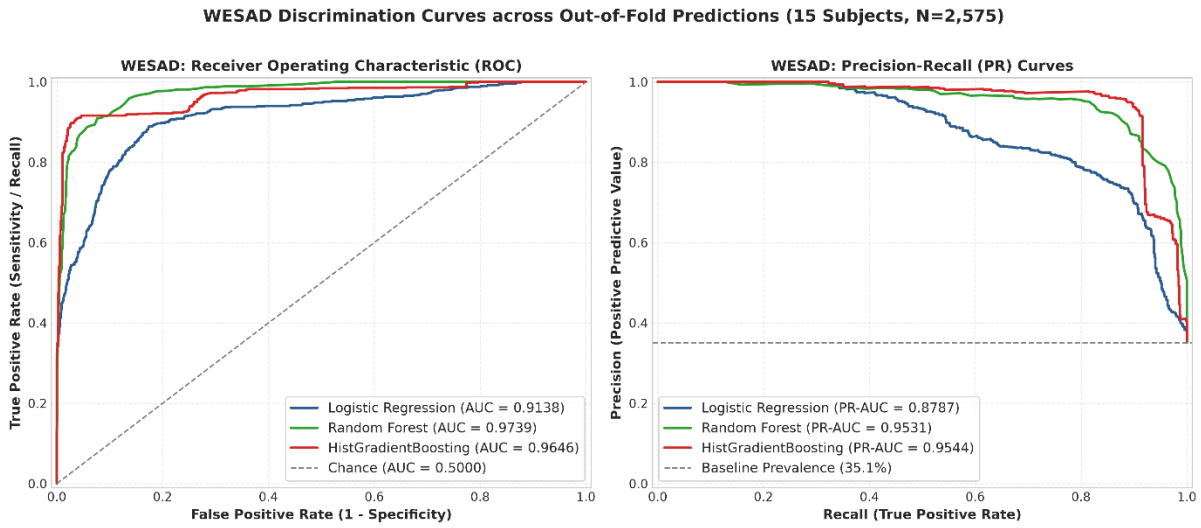
complexities of real physiological responses. Therefore, these results cannot be used as evidence of equivalent performance on actual human stress measurements.

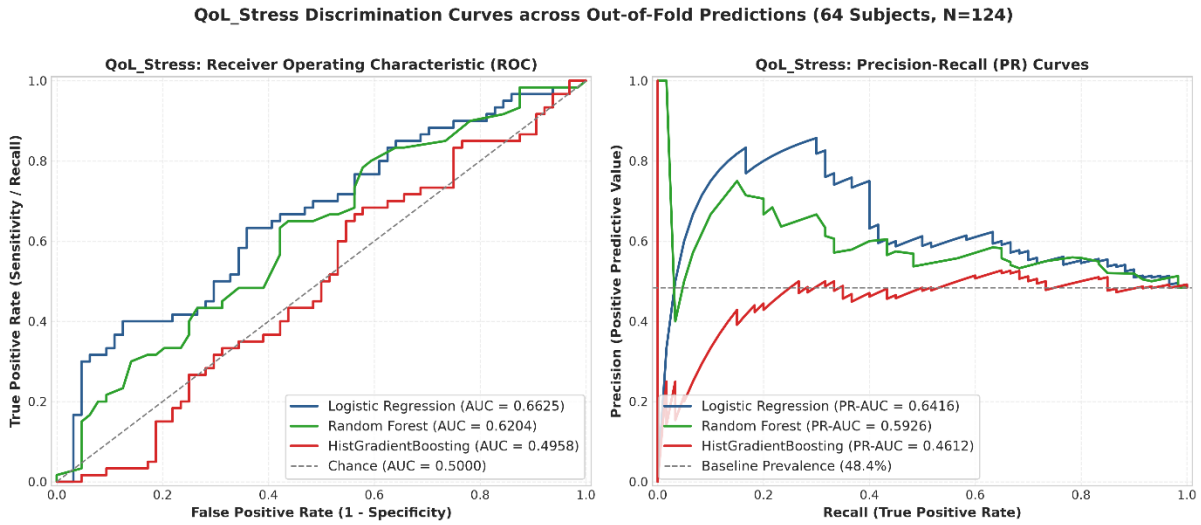
4.5 Comparative Discussion

The results demonstrate that model performance varied across the three datasets. HistGradientBoosting performed best on WESAD in terms of balanced accuracy and macro F1-score, while Logistic Regression performed best on QoL_Stress and the synthetic dataset for the reported metrics.

Table 4.6: Best-Performing Model by Dataset and Metric

Dataset	**Highest balanced accuracy**	**Highest macro F1-score**	**Highest ROC-AUC**
WESAD	HistGradientBoosting (0.9311)	HistGradientBoosting (0.9384)	Random Forest (0.9739)
QoL_Stress	Logistic Regression (0.6203)	Logistic Regression (0.6204)	Logistic Regression (0.6625)
Synthetic	Logistic Regression (0.8964)	Logistic Regression (0.8964)	Logistic Regression (0.9583)





The difference between the WESAD and QoL_Stress results highlights the importance of dataset characteristics and evaluation design. WESAD provided many more model observations through windowing, whereas QoL_Stress contained a smaller number of session-level observations. The datasets also differed in their recording protocols and physiological feature representations [1], [2].

The synthetic experiment produced strong results, but it represents a separate simulated setting. Its scores must not be combined with the empirical results or interpreted as evidence of real-world generalisation.

Overall, the experiments indicate that no single algorithm was consistently best across all datasets. Model selection should therefore consider the dataset, evaluation metrics, and the consequences of false positives and false negatives rather than relying on one score alone.

4.6 Chapter Summary

This chapter presented the results of Logistic Regression, Random Forest, and HistGradientBoosting on WESAD, QoL_Stress, and synthetic data. HistGradientBoosting achieved the highest balanced accuracy on WESAD, Logistic Regression performed best on QoL_Stress, and Logistic Regression also achieved the highest reported metrics on the synthetic dataset.

The findings demonstrate variation in model performance across different experimental settings. The human-dataset results provide the main evidence for evaluating physiological stress classification, while the synthetic experiment provides a separate assessment using simulated observations. The following chapter summarises the conclusions, limitations, and future scope of the project.

CHAPTER 5: CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This project investigated the use of machine learning for physiological stress classification using the WESAD and QoL_Stress datasets, along with a separate synthetic-data experiment. Logistic Regression, Random Forest, and HistGradientBoosting were evaluated using participant-independent Leave-One-Subject-Out validation.

The WESAD experiment produced the strongest classification results among the human-dataset experiments. HistGradientBoosting achieved a balanced accuracy of 0.9311 and a macro F1-score of 0.9384, while Random Forest achieved the highest ROC-AUC of 0.9739. These findings indicate that the extracted physiological features contained useful information for distinguishing baseline and stress conditions in WESAD.

Performance was lower on QoL_Stress. Logistic Regression achieved the highest balanced accuracy of 0.6203 and macro F1-score of 0.6204. This difference highlights the challenges of classifying stress using a smaller session-level dataset and reinforces the importance of evaluating models across different datasets rather than assuming that performance on one dataset will generalise to another.

In the synthetic-data experiment, Logistic Regression achieved the highest balanced accuracy of 0.8964 and macro F1-score of 0.8964. However, these results describe performance on simulated observations and should not be interpreted as evidence of equivalent performance on real human physiological data.

The project also demonstrated the importance of preventing participant-level data leakage, excluding target-derived predictors, and using multiple evaluation metrics. Overall, the findings show that machine learning can identify patterns associated with experimental stress conditions, but its performance depends on the dataset, feature representation, and evaluation setting. The results do not establish that the models are ready for real-world stress monitoring or clinical use.

5.2 Limitations

The project has several limitations.

1. Dataset size and composition: The two human datasets differ in participant count, recording protocols, and observation structure. The QoL_Stress dataset contains relatively few session-level observations, which limits the amount of data available for model training and evaluation.
2. Controlled experimental conditions: The datasets were collected under specific experimental conditions. Their results may not represent the complexity of stress responses during everyday activities.
3. Participant variability: Physiological responses differ across individuals. Although LOSO validation evaluates performance on held-out participants, it does not establish generalisation to every population, wearable device, or environment.
4. Differences between datasets: WESAD and QoL_Stress use different measurement protocols and feature representations. Their results should therefore be interpreted separately rather than as directly equivalent experiments.
5. Synthetic-data limitations: The synthetic dataset contains simulated observations rather than actual human measurements. Its results provide a separate modelling experiment but do not establish real-world predictive performance.

6. Model scope: The project evaluates three classification algorithms. Other algorithms, calibration techniques, and feature representations could produce different results and were outside the scope of the current comparison.

5.3 Future Scope

Future work can extend the project in several directions.

1. Larger and more diverse datasets: Additional participants, recording environments, and wearable devices could be included to examine whether the observed patterns generalise beyond the existing datasets.
2. Cross-dataset evaluation: Models trained on one dataset could be evaluated on another using carefully aligned features and compatible target definitions. This would provide a more direct assessment of cross-dataset generalisation.
3. Improved feature engineering: Additional physiological features and signal-processing methods could be investigated to determine whether they improve participant-independent classification.
4. Model calibration and uncertainty estimation: Future experiments could assess whether predicted probabilities are reliable and whether uncertain predictions should be flagged rather than treated as definitive classifications.
5. Real-time stress monitoring: The classification workflow could be extended into a prototype that processes incoming wearable measurements. Such a system would require additional evaluation for latency, signal quality, missing measurements, and performance under real-world conditions.
6. Explainability and personalised modelling: Feature-importance analysis and interpretable explanations could help investigate which physiological measurements influence model predictions. Personalised approaches could also be studied to account for differences in individual baselines.
7. Prospective evaluation: Before practical deployment, the models would require testing using independently collected data under realistic conditions, with appropriate privacy safeguards and clear communication that predictions are estimates rather than medical diagnoses.

5.4 Final Remarks

The project provides a comparative evaluation of three machine learning algorithms for physiological stress classification using two human datasets and a separate synthetic dataset. The findings demonstrate that the choice of model and the characteristics of the evaluation dataset influence predictive performance.

A key contribution of the workflow is its emphasis on participant-independent validation and leakage prevention. These measures provide a more meaningful assessment of how models may perform on participants whose observations were not used during training.

Although the results are promising for further research, additional validation is necessary before applying the models to everyday stress monitoring. Future work should prioritise generalisation, uncertainty estimation, and evaluation under realistic recording conditions.

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